

INSTITUTO SUPERIOR TÉCNICO

MASTERS IN MECHANICAL ENGINEERING

OPTIMAL CONTROL LABORATORY PROJECT - PHASE I

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Modeling and Simulation of a Quadcopter

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Introduction

This report presents the work developed in Phase I of the laboratory project for the Optimal Control course. The objective of this phase is to model and validate the dynamics of the Parrot Mambo Fly quadcopter, providing the foundation for the control design in subsequent phases.

The report is structured as follows. First, the nonlinear model of the quadcopter is implemented in Simulink based on the Newton-Euler equations of motion, and validated through a series of open-loop tests. Second, the model is linearized around the hovering equilibrium point, yielding four decoupled state-space subsystems, which are analyzed in terms of stability, controllability and observability, and validated against the nonlinear model. Third, a full-state feedback regulator based on pole placement is designed and tested. Finally, motor saturation is incorporated into the nonlinear model and its effect on closed-loop performance is assessed.

1 Nonlinear Model

The nonlinear model of the quadcopter is implemented in Simulink following the Newton-Euler equations of motion (Equation 2.1), with the generalized force $[F_3, M_1, M_2, M_3]^T$ as input and an additional port for external force perturbations in all three axes. The model computes the forward dynamics by numerically integrating the linear and angular accelerations (Equations 2.18 and 2.19) to obtain velocities and positions, while the orientation matrix $\mathbf{R}_n$ is updated at each timestep via the matrix exponential method (Equation 2.22). The overall Simulink implementation is illustrated in Figures 1 and 2.

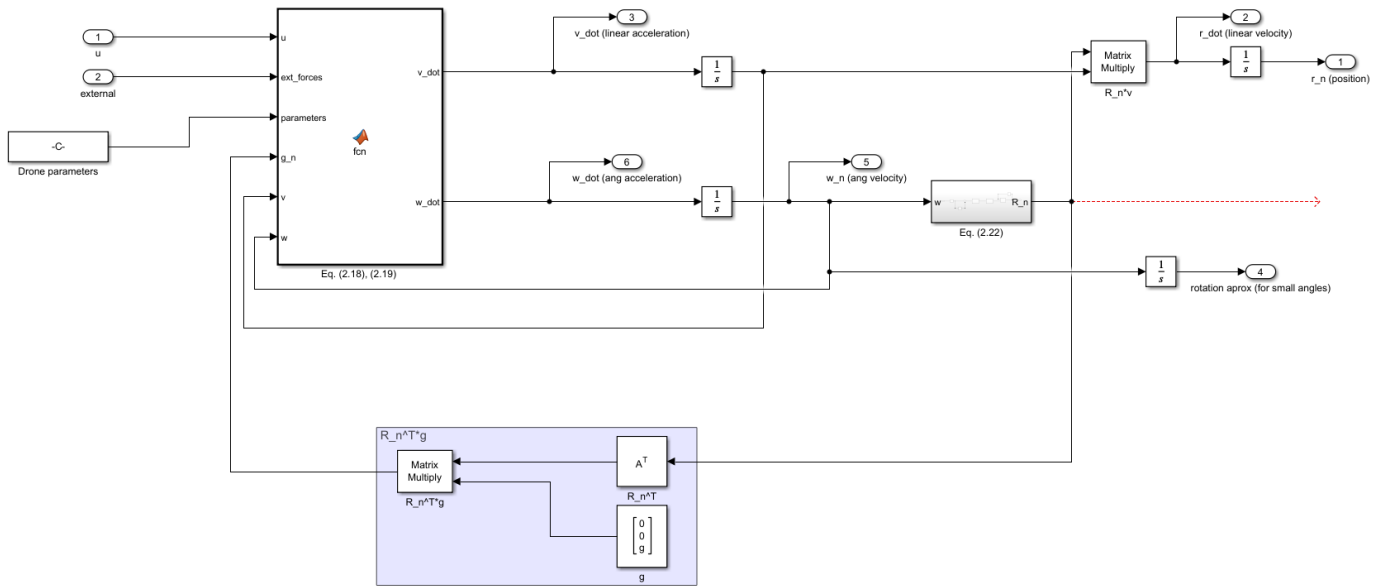


Figure 1: Simulink block diagram of the nonlinear quadcopter model, implementing Equations 2.18 and 2.19 with orientation update via matrix exponential.

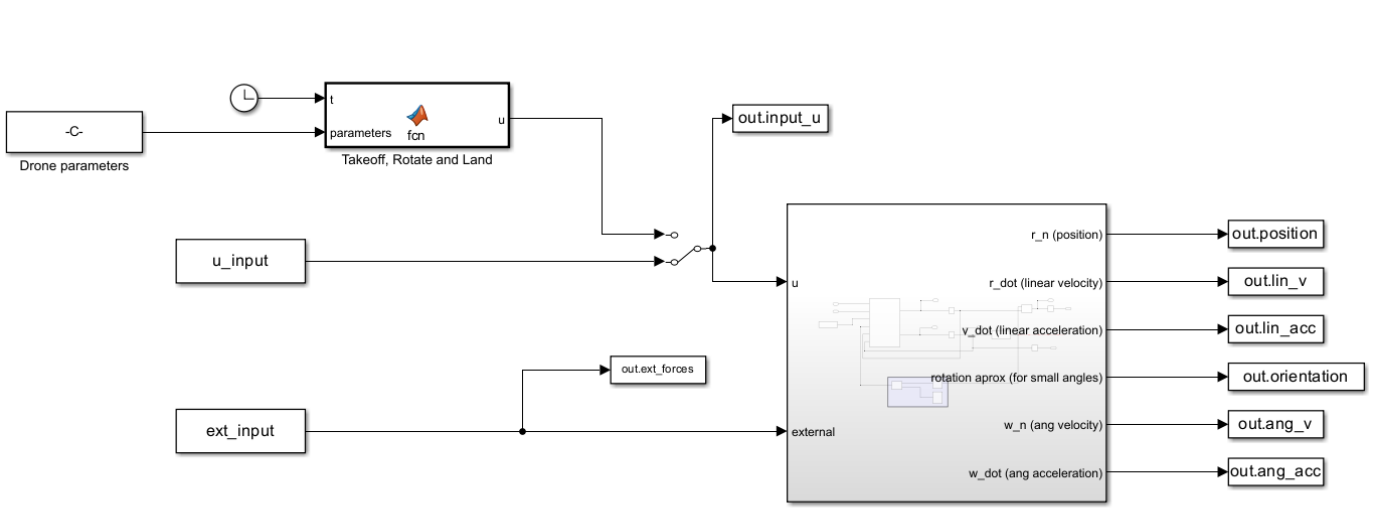


Figure 2: Simulink input structure for the nonlinear model, showing how the generalized forces and external perturbations are fed into the model for the validation tests.

1.1 System States and Input Variables

The state vector is described by 12 variables, capturing position, orientation, linear velocity and angular velocity along all three axes:

$$x = \begin{bmatrix} r_i \\ \dot{r}_i \\ \theta_i \\ \dot{\theta}_i \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ \dot{r}_1 \\ \dot{r}_2 \\ \dot{r}_3 \\ \theta_1 \\ \theta_2 \\ \theta_3 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \quad (1)$$

The input vector consists of a vertical thrust force and moments about each body axis:

$$u = \begin{bmatrix} F_3 \\ M_1 \\ M_2 \\ M_3 \end{bmatrix} \quad (2)$$

Note that $F_1 = F_2 = 0$ always, since propeller thrust is aligned with body axis Z_n .

1.2 Hover Thrust Determination

At hover, all velocities and accelerations are zero. From Equation 2.29, the vertical force must balance gravity:

$$F_3 = -m_n g_3 = -m_n g \approx -0.618 \text{ N} \quad (3)$$

The negative sign reflects the body-frame convention: Z_n points downward, so thrust must act in the negative Z_n direction to oppose gravity. With $F_3 = -mg$ applied and zero initial conditions, all states remain identically zero throughout the simulation, confirming hover equilibrium.

Q2 - Hover — States

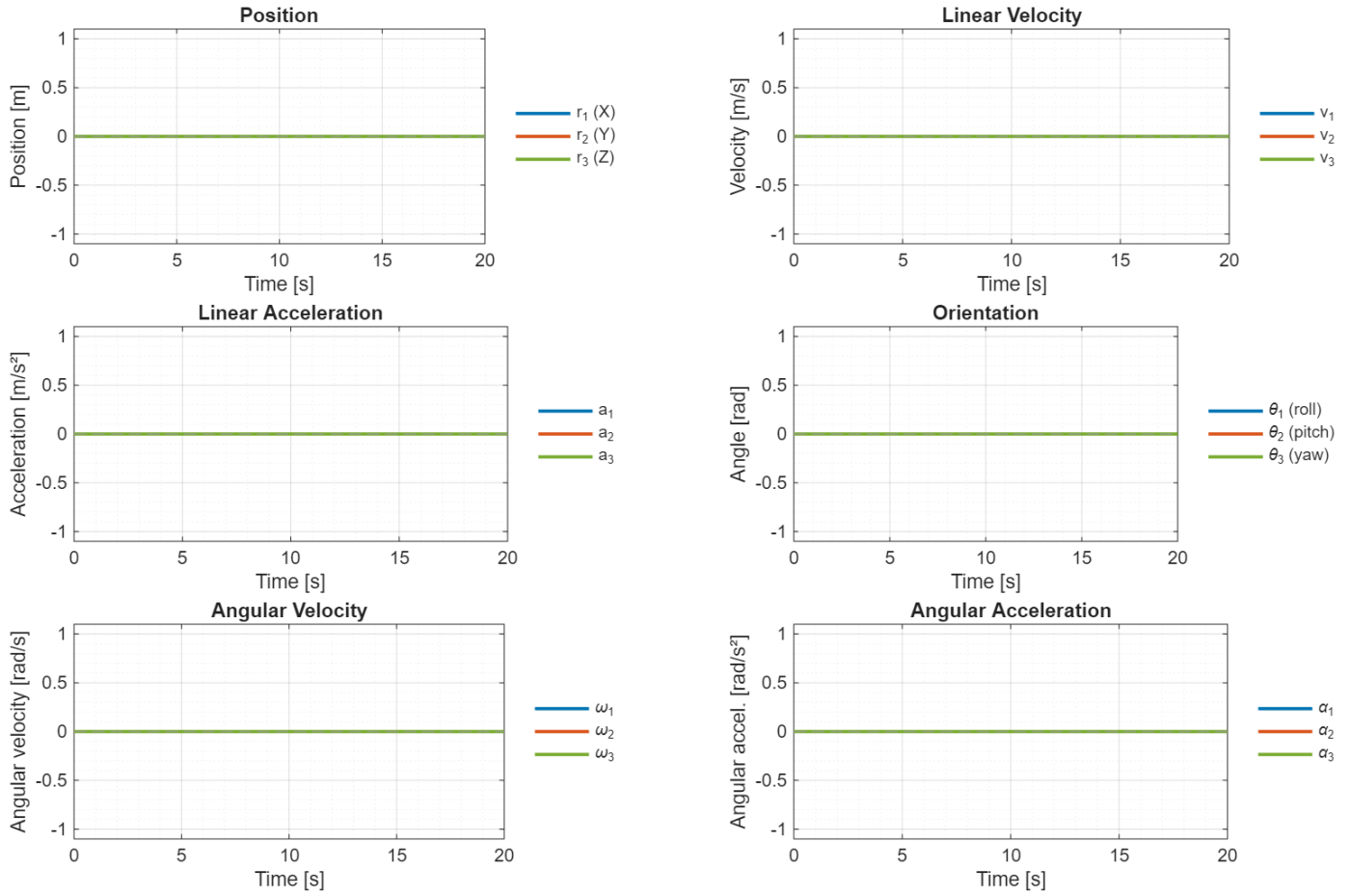


Figure 3: Q2 — Hover: all states remain at zero confirming equilibrium.

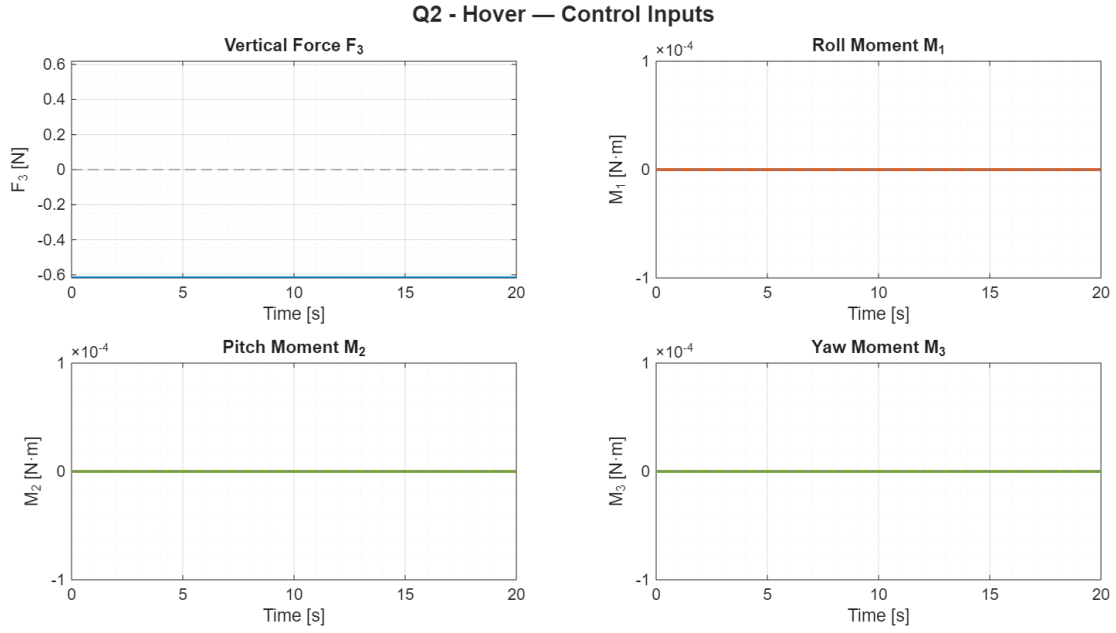


Figure 4: Q2 — Hover: constant vertical force $F_3 = -mg$, all moments zero.

1.3 Step Response in Vertical Force

A step from $F_3 = 0$ to $F_3 = -1.5$ N is applied at $t = 2$ s. Before the step, the drone is in free fall under gravity. After the step, the net vertical acceleration becomes:

$$\ddot{r}_3 = \frac{F_3}{m} + g = \frac{-1.5}{0.063} + 9.81 \approx -14 \text{ m/s}^2 \quad (4)$$

The net upward acceleration decelerates the falling drone, which then reverses direction. All other axes remain unaffected, confirming that vertical dynamics are fully decoupled from the remaining states. This is consistent with the double-integrator transfer function $H_1(s) = 1/(m_n s^2)$.

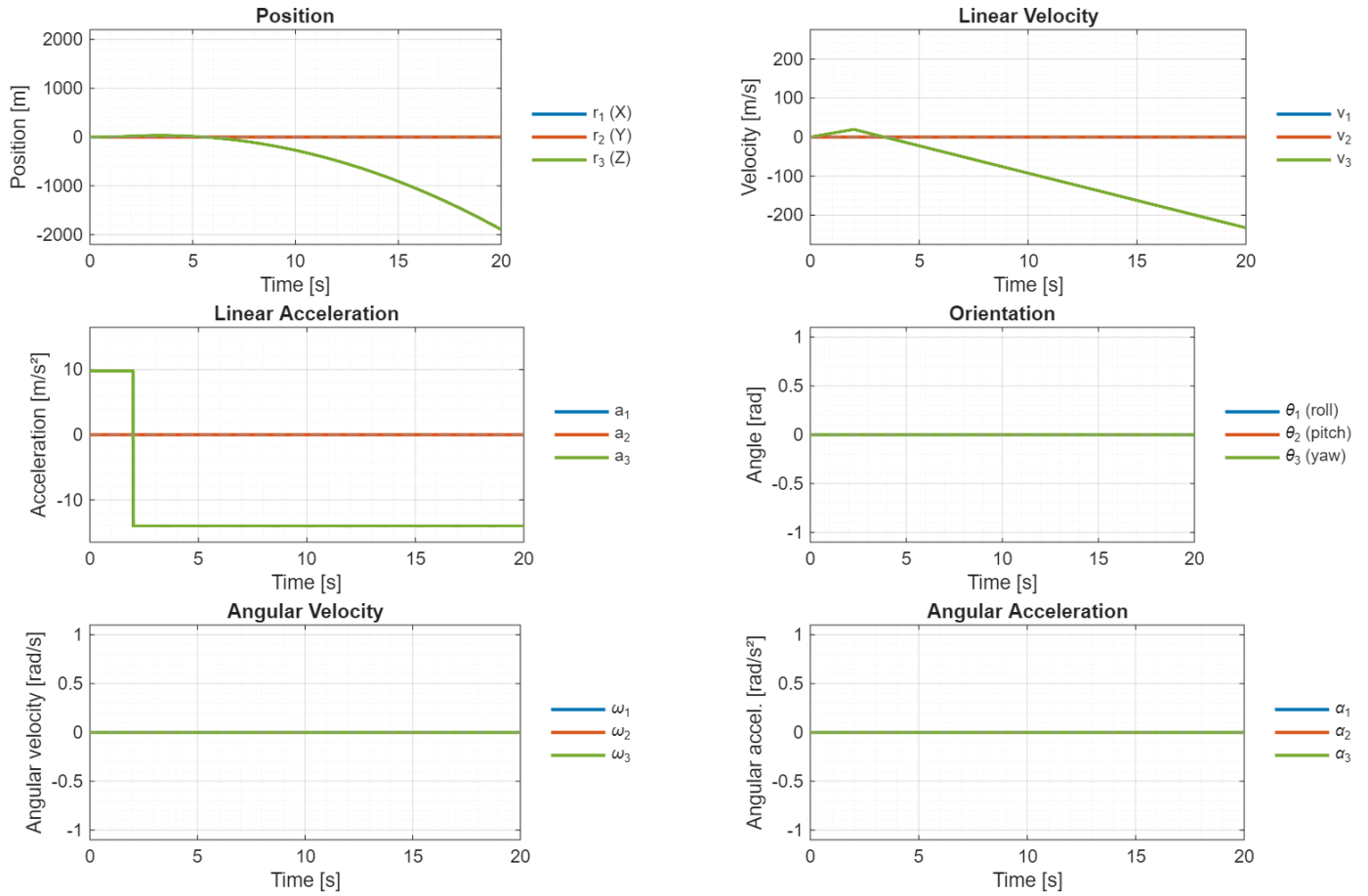
Q3 - Step in F_3 — States


Figure 5: Q3 — Step in F_3 : free fall before step, net upward acceleration after.

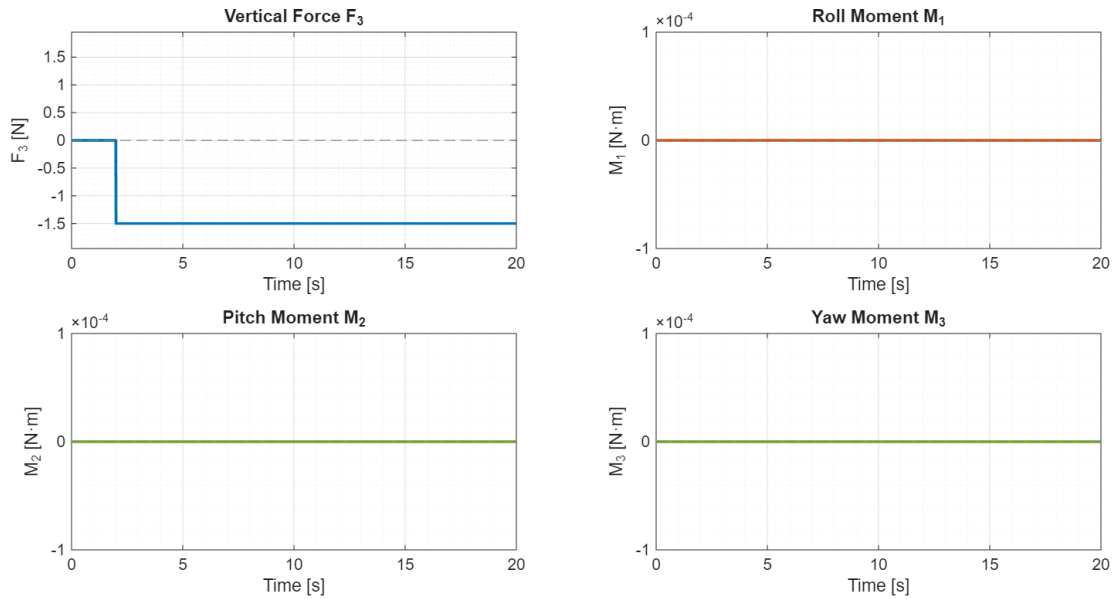
Q3 - Step in F_3 — Control Inputs


Figure 6: Q3 — Step in F_3 : input steps from 0 to -1.5 N at $t = 2$ s.

1.4 Yaw Response to a Step Moment

While hovering ($F_3 = -mg$), a step $M_3 = 0.001 \text{ N}\cdot\text{m}$ is applied at $t = 2 \text{ s}$. The expected yaw angular acceleration is:

$$\ddot{\theta}_3 = \frac{M_3}{J_{33}} = \frac{0.001}{3 \times 10^{-5}} \approx 33.3 \text{ rad/s}^2 \quad (5)$$

As expected, θ_3 grows as a parabola, ω_3 ramps linearly, and α_3 is constant after the step. All translational states and the remaining rotational axes remain at zero, confirming that yaw is fully decoupled. This is consistent with $H_4(s) = 1/(J_{33}s^2)$.

Q4 - Step in M3 (Yaw) — States

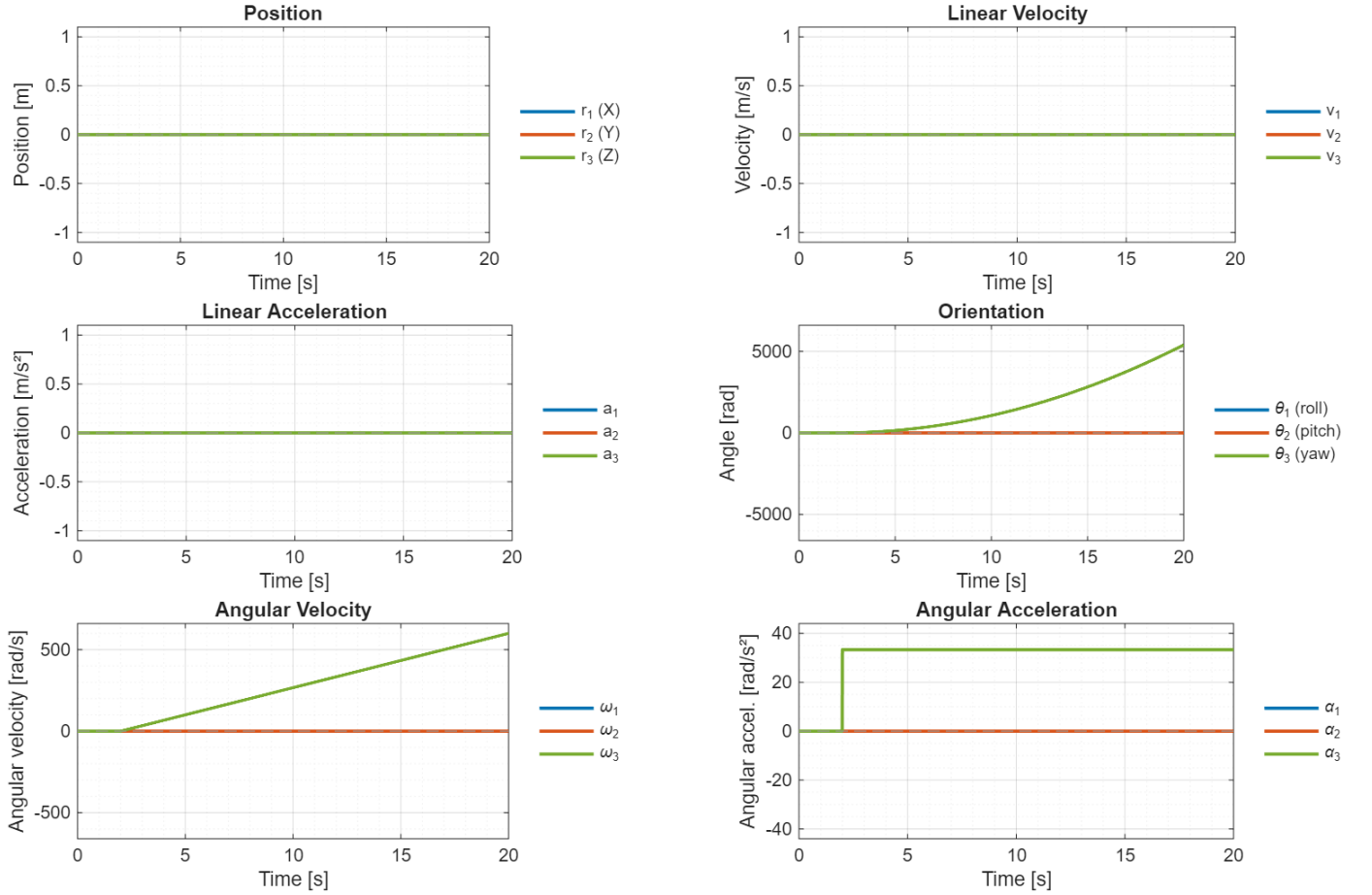


Figure 7: Q4 — Step in M_3 : yaw grows as double integrator, all other states unaffected.

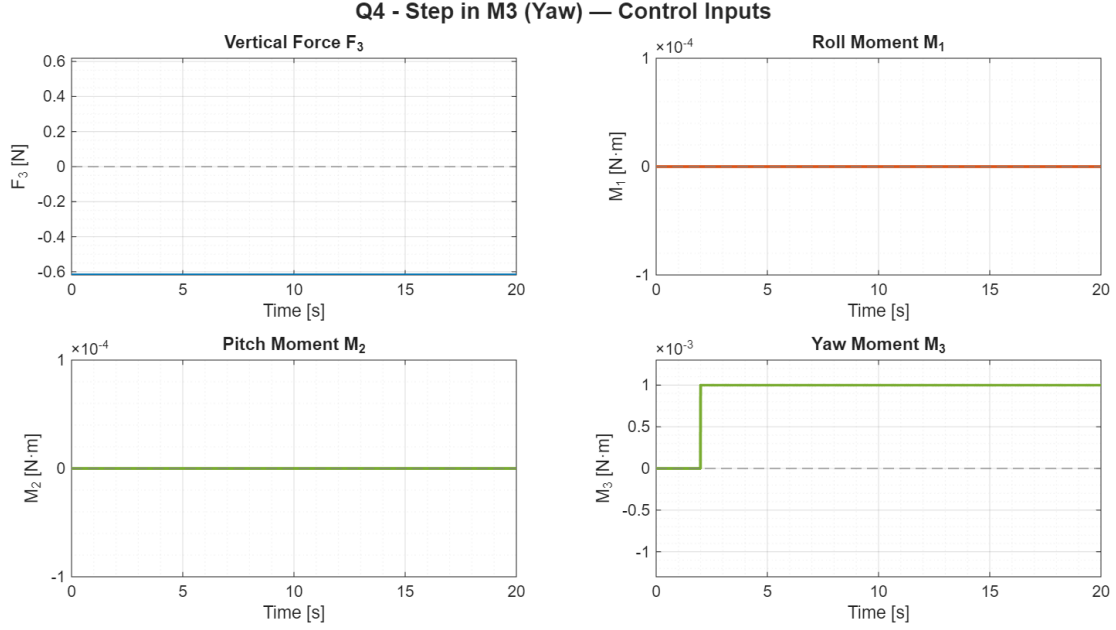


Figure 8: Q4 — Step in M_3 : hover force maintained, yaw moment step applied.

1.5 Pitch Moment Step Response Analysis

Although the question does not explicitly specify whether the moment should be applied with or without hover thrust, $F_3 = -mg$ is maintained throughout this test. This choice isolates the pitch coupling effect more clearly: without hover thrust, r_3 would drift simultaneously due to gravity, mixing two independent effects and obscuring the interpretation. A step $M_2 = 5 \times 10^{-5}$ N·m is applied at $t = 2$ s while hovering. The expected pitch angular acceleration is:

$$\ddot{\theta}_2 = \frac{M_2}{J_{22}} = \frac{5 \times 10^{-5}}{7.1691 \times 10^{-5}} \approx 0.698 \text{ rad/s}^2 \quad (6)$$

As θ_2 grows, gravity projects onto the longitudinal axis via $\ddot{r}_1 = -g\theta_2$, causing r_1 to drift in the negative X direction. All remaining axes (r_2, θ_1, θ_3) stay at zero, confirming that pitch is decoupled from lateral and yaw dynamics. Although hover thrust $F_3 = -mg$ is maintained, as θ_2 grows the thrust vector tilts with the body frame away from the vertical. Its vertical component, now $F_3 \cos(\theta_2)$, is no longer sufficient to balance gravity, causing the drone to descend and r_3 to grow parabolically. This is a purely nonlinear effect — the linearized model assumes small angles where $\cos(\theta_2) \approx 1$, and this descent would not be observed there.

The linear accelerations exhibit growing oscillatory behaviour over time, a consequence of the nonlinear Coriolis terms $v_i \times \omega_j$ in Equation 2.18. As $\dot{r}_1, \dot{r}_3$ and ω_2 grow continuously, their cross products become significant and introduce coupling between axes. This behaviour is purely a nonlinear effect and does not appear in the linearized model, where these quadratic terms are neglected.

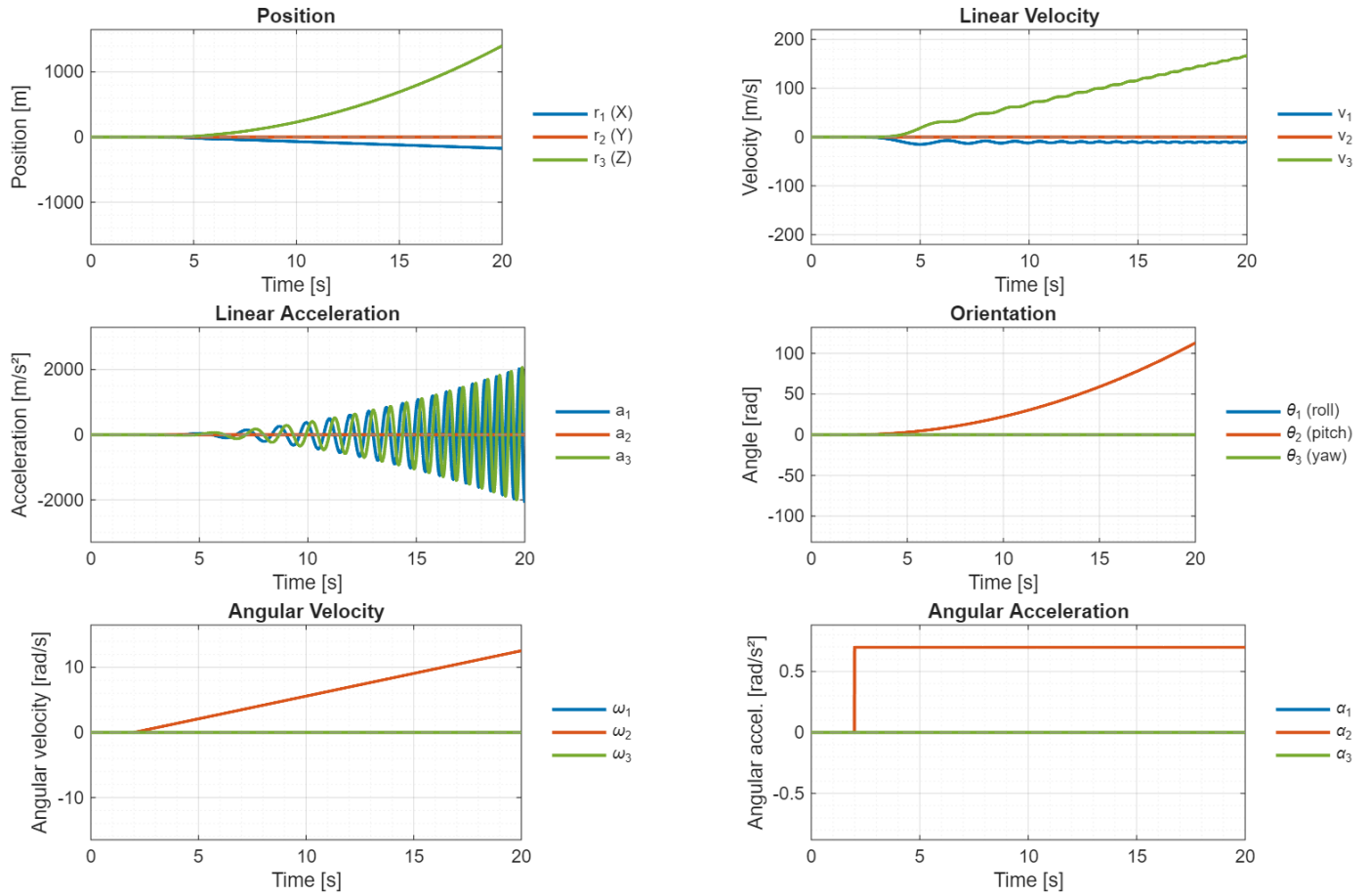
Q5 - Step in M_2 (Pitch) — States


Figure 9: Q5 — Step in M_2 : pitch grows as double integrator, coupling drives r_1 drift.

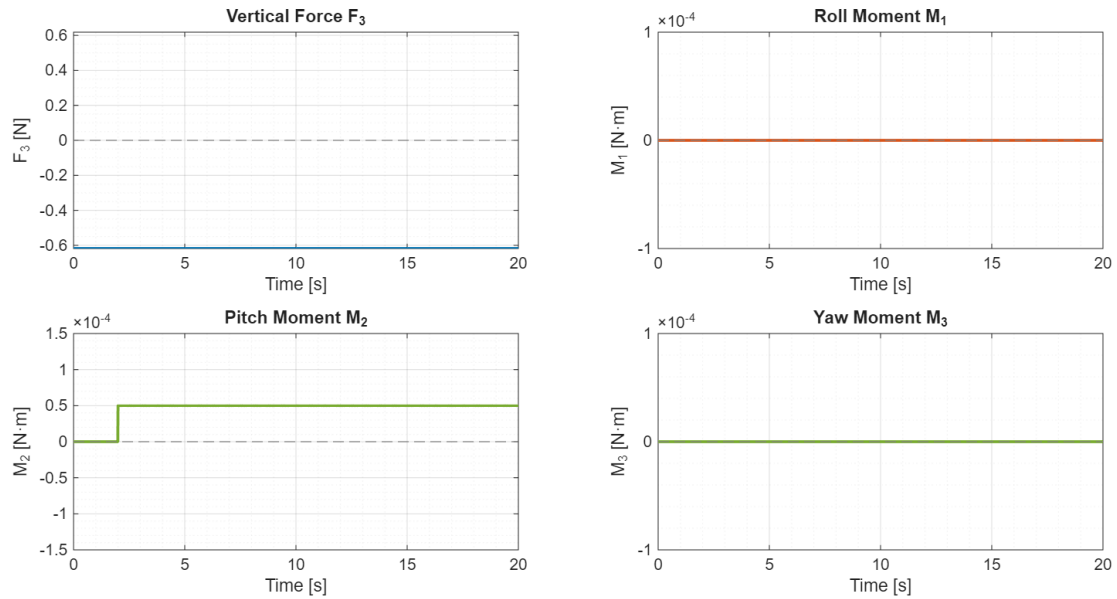
Q5 - Step in M_2 (Pitch) — Control Inputs


Figure 10: Q5 — Step in M_2 : pitch moment step from zero.

1.6 Lateral Perturbation During Hover

While hovering ($F_3 = -mg$), a brief lateral force pulse $F_{2,\text{ext}} = 0.1 \text{ N}$ is applied along the body Y axis for 1 second ($2 \text{ s} \leq t < 3 \text{ s}$). The expected lateral acceleration during the pulse is:

$$\ddot{r}_2 = \frac{F_{2,\text{ext}}}{m} = \frac{0.1}{0.063} \approx 1.587 \text{ m/s}^2 \quad (7)$$

During the pulse, the constant acceleration causes r_2 to grow parabolically, while $\dot{r}_2$ ramps linearly. Once the pulse ends at $t = 3 \text{ s}$, the external force disappears but the drone retains the acquired lateral velocity $\dot{r}_2 \approx 1.587 \text{ m/s}$, drifting indefinitely in the Y direction with no restoring force and r_2 now growing linearly. All remaining states ($r_1, r_3, \theta_1, \theta_2, \theta_3$) remain at zero throughout, confirming that the lateral dynamics are fully decoupled from all other axes.

This result highlights the *marginal stability* of the uncontrolled quadcopter: a brief bounded disturbance produces *unbounded* positional drift, with no tendency to return to the original position. This directly motivates the need for a closed-loop controller.

Q6 - Lateral Perturbation — States

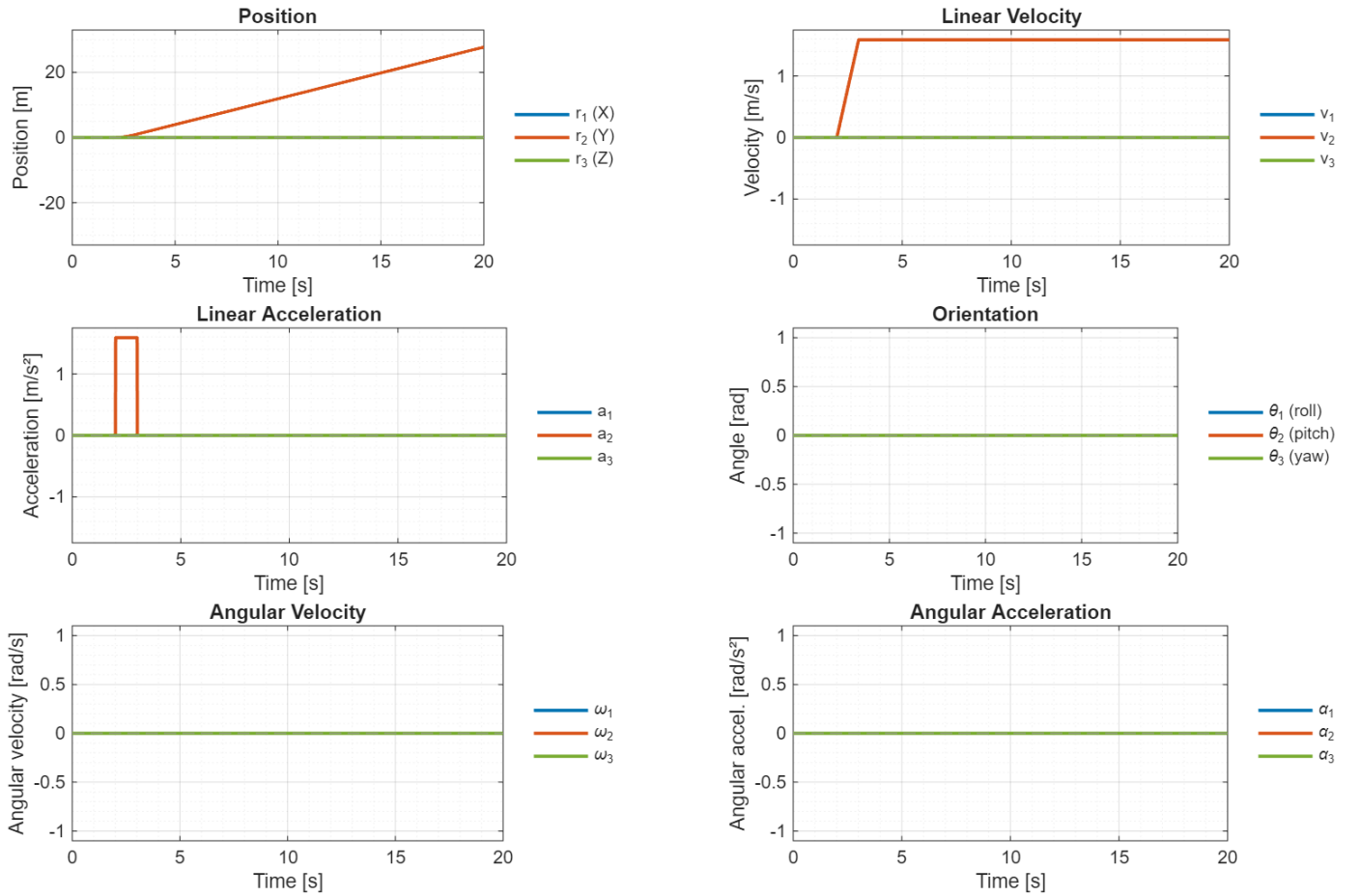


Figure 11: Q6 — Lateral perturbation: r_2 drifts linearly after the pulse, $\dot{r}_2$ reaches a constant value, all other states unaffected.

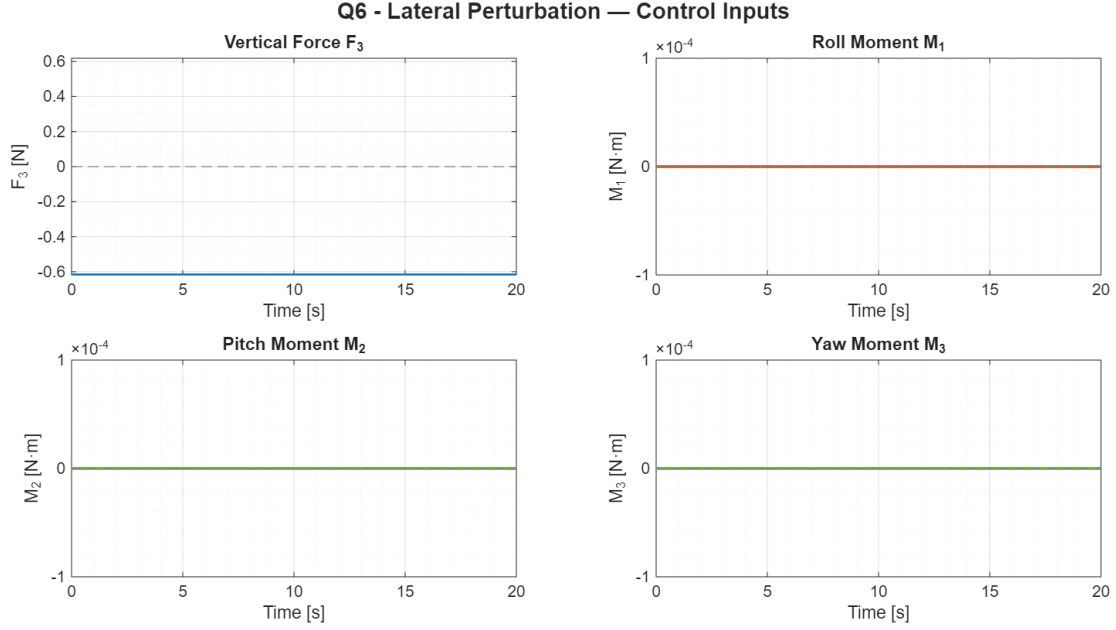


Figure 12: Q6 — Control inputs: hover force $F_3 = -mg$ maintained, all moments zero.

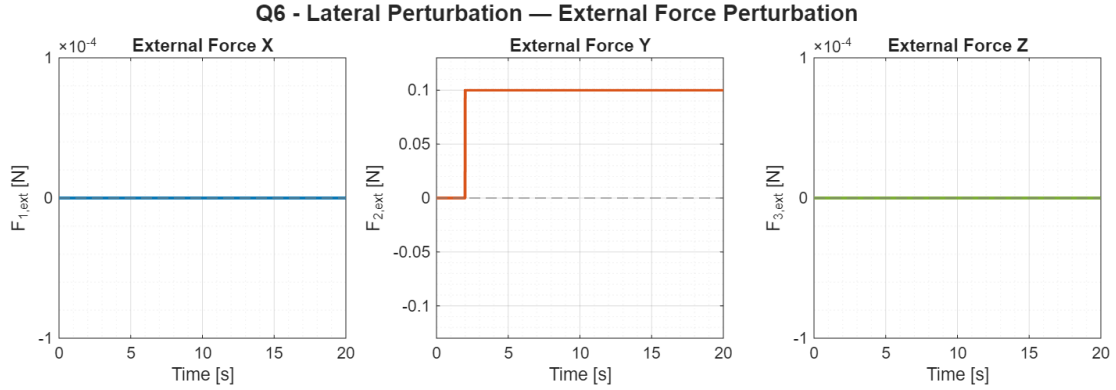


Figure 13: Q6 — External force: lateral pulse $F_{2,ext} = 0.1$ N applied for 1 second.

1.7 Takeoff, Rotate and Land Trajectory Validation

The desired trajectory is defined by piecewise constant accelerations, with the generalized forces computed via Equation 2.28 of the Quadcopter Dynamics guide:

$$\begin{bmatrix} F_3 \\ M_1 \\ M_2 \\ M_3 \end{bmatrix}_d = \begin{bmatrix} m_n \ddot{q}(\alpha_{TO}, t_{TO}, t) + m_n \ddot{q}(\alpha_{Land}, t_{Land}, t - t_{TO} - t_{Rot}) - m_n g_3 \\ 0 \\ 0 \\ J_{33} \ddot{q}(\alpha_{Rot}, t_{Rot}, t - t_{TO}) \end{bmatrix} \quad (8)$$

The simulation uses $t_{TO} = 5$ s, $t_{Rot} = 10$ s, $t_{Land} = 5$ s, with a total displacement of -10 cm along Z_n and a π rotation about Z_n . The results confirm that r_3 follows the expected trapezoidal velocity profile and θ_3 reaches π rad, while all remaining states stay at zero throughout.

Q7 - Takeoff Rotate Land — States

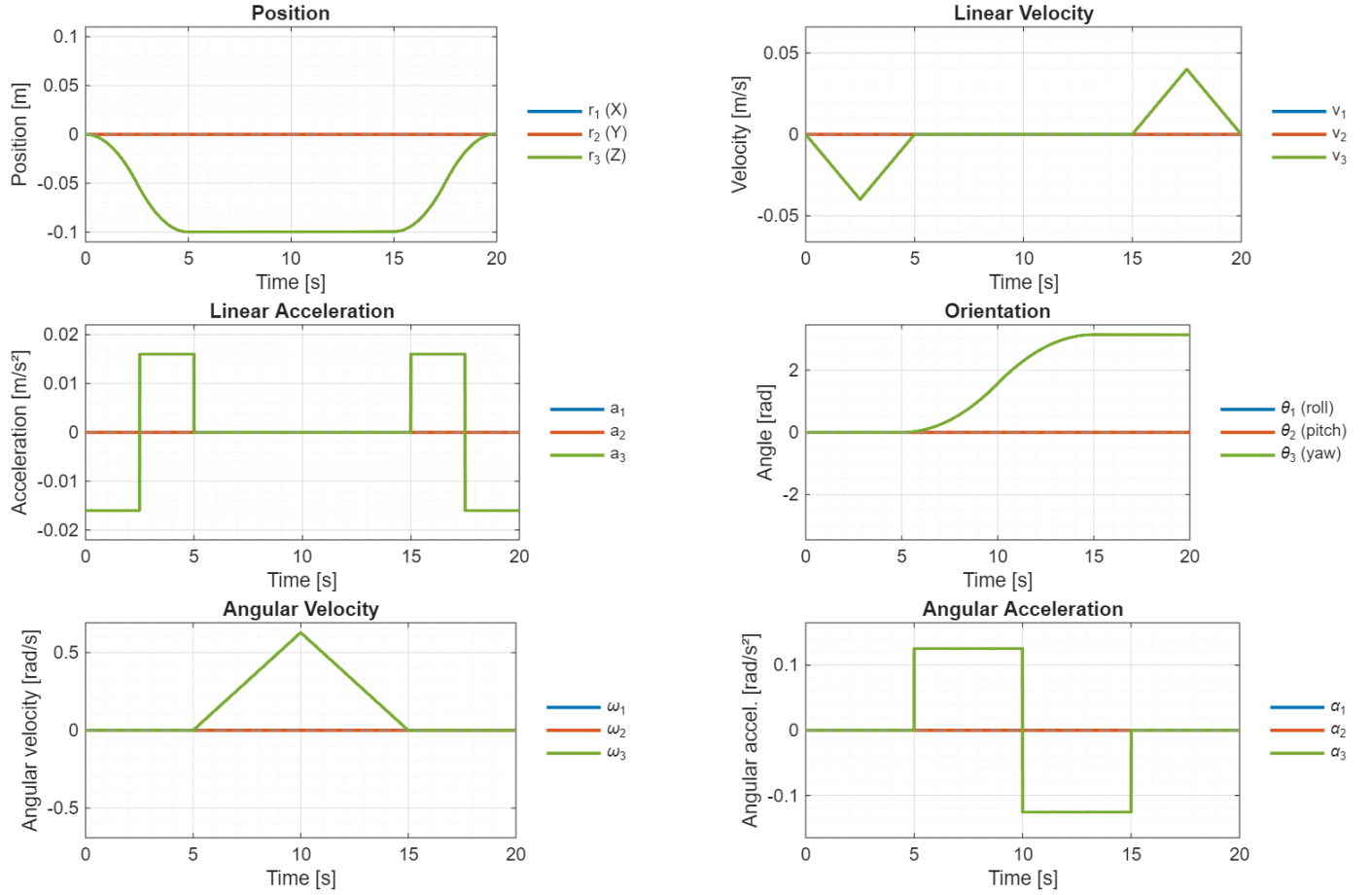


Figure 14: Q7 — Takeoff, rotate and land: position and orientation follow desired trajectory.

The reconstructed control inputs are shown in Figure 15. The vertical force F_3 varies slightly around the hover value ($F_3 = -mg \approx -0.618$ N), increasing in magnitude during takeoff and decreasing during landing, consistent with the required upward and downward accelerations. The moments M_1 and M_2 remain identically zero throughout, confirming that no roll or pitch is induced. The yaw moment M_3 follows a piecewise constant profile during the rotation phase, positive during the acceleration half and negative during the deceleration half, producing the desired π rad rotation about the vertical axis.

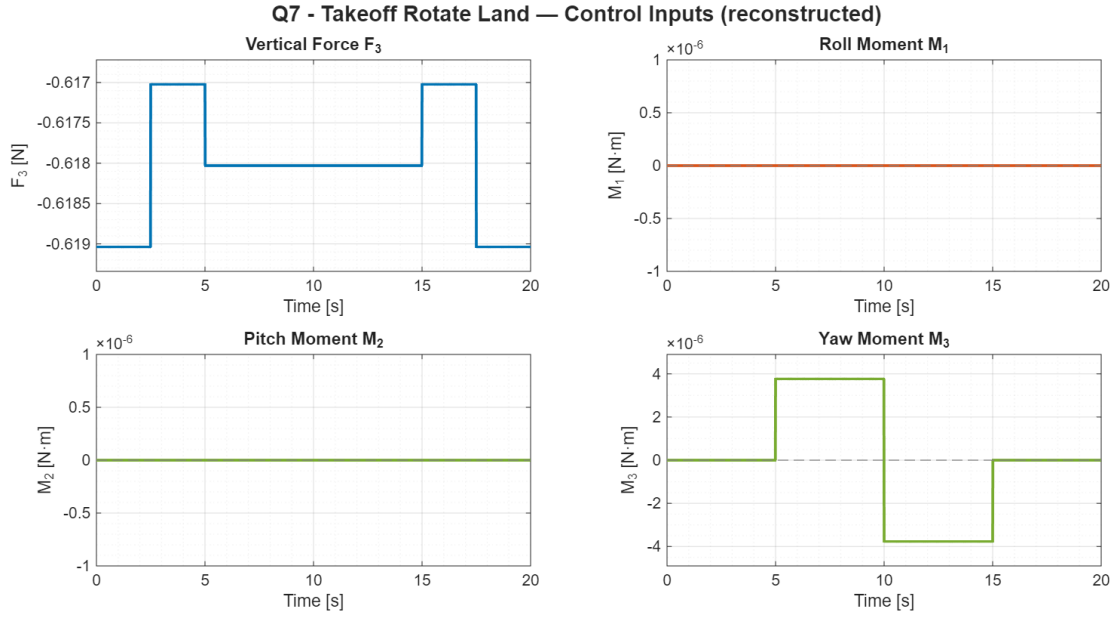


Figure 15: Q7 — Takeoff, rotate and land: reconstructed control inputs F_3 and M_3 .

2 Linearized Model

The nonlinear quadcopter model is linearized around the hovering equilibrium point, where the drone is stationary with zero velocities and zero angular rates. Following the procedure described in Section 2.4 of the Quadcopter Dynamics guide, small perturbations are introduced around this operating point and the equations of motion are expanded to first order, discarding all quadratic and higher-order terms. The key nonlinearities eliminated in this process are the Coriolis terms $v_i \times \omega_j$, the gyroscopic terms $\omega_i \times J\omega_j$, and the rotation matrix $\mathbf{R}_n$ in the gravity projection. The result is a linear state-space representation consisting of four fully decoupled subsystems, each governed by a double-integrator structure. The Simulink implementation is shown in Figures 16 and 17.

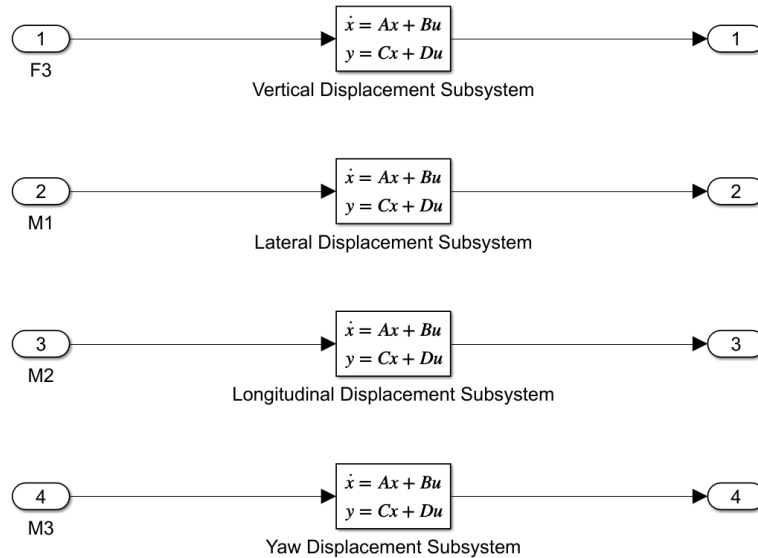


Figure 16: Simulink block diagram of the linearized quadcopter model, implementing the four decoupled state-space subsystems.

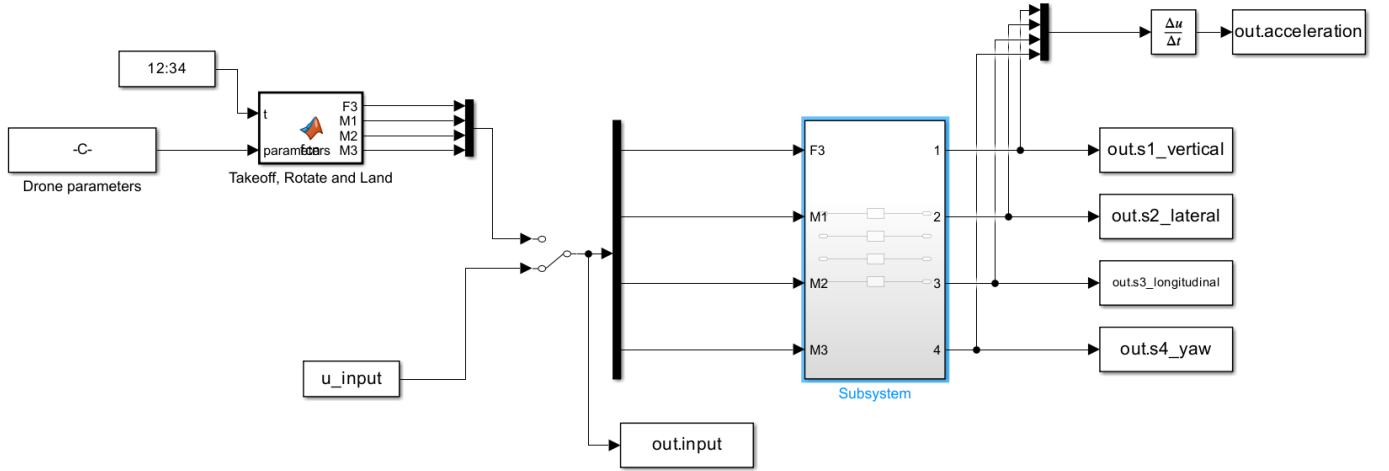


Figure 17: Simulink input structure for the linearized model, showing how the perturbation inputs $[F'_3, M'_1, M'_2, M'_3]^\top$ are fed into each subsystem for the validation tests.

2.1 State-Space Representation of Each Subsystem

The input variables do not exactly match those of the nonlinear model. Instead, they represent the perturbations of the vertical force (rather than its full value) and the perturbations of the moments around each axis (rather than their full values). Note that since the equilibrium moments are zero, $M'_i = M_i$ exactly; only F'_3 differs numerically from its nonlinear counterpart, as $F'_3 = F_3 - F_{3,eq} = F_3 + mg$. The outputs consist of the angular and linear displacements and velocities.

$$\mathbf{x} = [r_3 \ \dot{r}_3 \ r_2 \ \dot{r}_2 \ \theta'_1 \ \dot{\theta}'_1 \ r_1 \ \dot{r}_1 \ \theta'_2 \ \dot{\theta}'_2 \ \theta'_3 \ \dot{\theta}'_3]^\top \quad (9)$$

$$\mathbf{u} = [F'_3 \ M'_1 \ M'_2 \ M'_3]^\top \quad (10)$$

The full system follows the state-space representation:

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \quad (11)$$

$$\mathbf{y} = \mathbf{Cx} \quad (12)$$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & g & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -g & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ \hline \frac{1}{m_a} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{J_{11}} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J_{22}} & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{J_{33}} \end{bmatrix} \mathbf{u}$$

Figure 18: Full system state-space matrices **A** and **B**.

Matrix **A** is divided into four submatrices: A_1 , A_2 , A_3 , and A_4 . Similarly, matrix **B** is partitioned into B_1 , B_2 , B_3 , and B_4 . Matrix **C** = $\mathbf{I}_{12 \times 12}$ is an identity matrix, segmented into four identity submatrices (C_1 , C_2 , C_3 , C_4) to match the dimensions of each subsystem's state-space model, meaning all states are directly measured as outputs. The linearized system consists of four fully decoupled subsystems, each actuated by a single control input. Their individual state-space matrices are summarized in Table 1.

Table 1: State-space representation of each decoupled subsystem.

Subsystem	A	$\mathbf{x}$	$\mathbf{u}$	B
1	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} r_3 \\ \dot{r}_3 \end{bmatrix}$	F'_3	$\begin{bmatrix} 0 \\ \frac{1}{m_a} \end{bmatrix}$
2	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} r_2 \\ \theta'_1 \\ \dot{r}_2 \\ \dot{\theta}'_1 \end{bmatrix}$	M'_1	$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_{11}} \end{bmatrix}$
3	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -g & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} r_1 \\ \theta'_2 \\ \dot{r}_1 \\ \dot{\theta}'_2 \end{bmatrix}$	M'_2	$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_{22}} \end{bmatrix}$
4	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \theta'_3 \\ \dot{\theta}'_3 \end{bmatrix}$	M'_3	$\begin{bmatrix} 0 \\ \frac{1}{J_{33}} \end{bmatrix}$

Each subsystem governs a distinct motion channel of the quadcopter: Subsystem 1 describes the *vertical dynamics*, driven by the thrust perturbation F'_3 ; Subsystem 2 captures the coupled *lateral-roll dynamics*, where a roll moment M'_1 induces both roll rotation and lateral translation; Subsystem 3 captures the coupled *longitudinal-pitch dynamics*, where a pitch moment M'_2 induces both pitch rotation and longitudinal translation; and Subsystem 4 describes the *yaw dynamics*, driven solely by M'_3 with no translational coupling.

2.2 Stability, Controllability and Observability Analysis

The state-space matrices of the full system are block-diagonal, with each block corresponding to one of the four decoupled subsystems. Therefore, analyzing the properties of each subsystem individually is equivalent to analyzing the full system.

Stability

A continuous-time system is stable if all of its poles lie in the open left half of the s -plane. If poles lie on the imaginary axis, they must be simple; otherwise the system is unstable.

$$|s\mathbf{I} - \mathbf{A}| = 0 \quad (13)$$

For subsystem 1:

$$|s\mathbf{I} - \mathbf{A}_1| = \begin{vmatrix} s & -1 \\ 0 & s \end{vmatrix} = s^2 \quad (14)$$

For subsystem 2:

$$|s\mathbf{I} - \mathbf{A}_2| = \begin{vmatrix} s & 0 & -1 & 0 \\ 0 & s & 0 & -1 \\ 0 & -g & s & 0 \\ 0 & 0 & 0 & s \end{vmatrix} = s^4 \quad (15)$$

For subsystem 3:

$$|s\mathbf{I} - \mathbf{A}_3| = \begin{vmatrix} s & 0 & -1 & 0 \\ 0 & s & 0 & -1 \\ 0 & g & s & 0 \\ 0 & 0 & 0 & s \end{vmatrix} = s^4 \quad (16)$$

For subsystem 4:

$$|s\mathbf{I} - \mathbf{A}_4| = \begin{vmatrix} s & -1 \\ 0 & s \end{vmatrix} = s^2 \quad (17)$$

All poles are located at the origin of the s -plane with multiplicity greater than 1. Therefore, the system is **unstable**, as poles on the imaginary axis with multiplicity greater than 1 cause the free response to grow polynomially with time. This is consistent with the double-integrator behavior observed in the nonlinear model validation, where bounded inputs produced unbounded positional drift.

Controllability

A system is fully controllable if the controllability matrix $\mathcal{C}$ has rank equal to n , the number of state variables.

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] \quad (18)$$

For subsystem 1:

$$\mathcal{C}_1 = \begin{bmatrix} 0 & \frac{1}{m_n} \\ \frac{1}{m_n} & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{C}_1) = 2 = n_1 \quad (19)$$

For subsystem 2:

$$\mathcal{C}_2 = \begin{bmatrix} 0 & 0 & 0 & \frac{g}{J_{11}} \\ 0 & \frac{1}{J_{11}} & 0 & 0 \\ 0 & 0 & \frac{g}{J_{11}} & 0 \\ \frac{1}{J_{11}} & 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{C}_2) = 4 = n_2 \quad (20)$$

For subsystem 3:

$$\mathcal{C}_3 = \begin{bmatrix} 0 & 0 & 0 & -\frac{g}{J_{22}} \\ 0 & \frac{1}{J_{22}} & 0 & 0 \\ 0 & 0 & -\frac{g}{J_{22}} & 0 \\ \frac{1}{J_{22}} & 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{C}_3) = 4 = n_3 \quad (21)$$

For subsystem 4:

$$\mathcal{C}_4 = \begin{bmatrix} 0 & \frac{1}{J_{33}} \\ \frac{1}{J_{33}} & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{C}_4) = 2 = n_4 \quad (22)$$

Every subsystem's controllability matrix is full rank, and therefore the overall system controllability matrix is also full rank ($\text{rank}(\mathcal{C}) = 12$). The system is **fully controllable**, meaning it is possible to drive the state to any desired configuration using appropriate inputs.

Observability

A system is fully observable if the observability matrix $\mathcal{O}$ has rank equal to n .

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \quad (23)$$

For subsystem 1:

$$\mathcal{O}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{O}_1) = 2 = n_1 \quad (24)$$

For subsystem 2:

$$\mathcal{O}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{O}_2) = 4 = n_2 \quad (25)$$

For subsystem 3:

$$\mathcal{O}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -g & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -g & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{O}_3) = 4 = n_3 \quad (26)$$

For subsystem 4:

$$\mathcal{O}_4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{rank}(\mathcal{O}_4) = 2 = n_4 \quad (27)$$

Since every subsystem's observability matrix is full rank, the overall system is **fully observable** ($\text{rank}(\mathcal{O}) = 12$). Given that $\mathbf{C} = \mathbf{I}_{12 \times 12}$, all states are directly measured, so a state estimator is not required at this stage.

2.3 Hover Thrust in the Linearized Model

Unlike the nonlinear model, which requires a constant vertical force $F_3 = -mg \approx -0.618$ N to maintain hover, the linearized model does not require any additional input. This is because the linearization is performed around the hovering equilibrium point, so the gravitational compensation is already embedded in the equilibrium condition itself. Consequently, the perturbation input $F'_3 = 0$ is sufficient to keep the drone stationary — in contrast to the nonlinear model where $F_3 = -mg$ must be explicitly applied at all times.

As illustrated in Figures 19 and 20, the system receives no vertical force input yet exhibits neither angular nor linear movement, as expected.

Q10 - Hover (Linear) — States

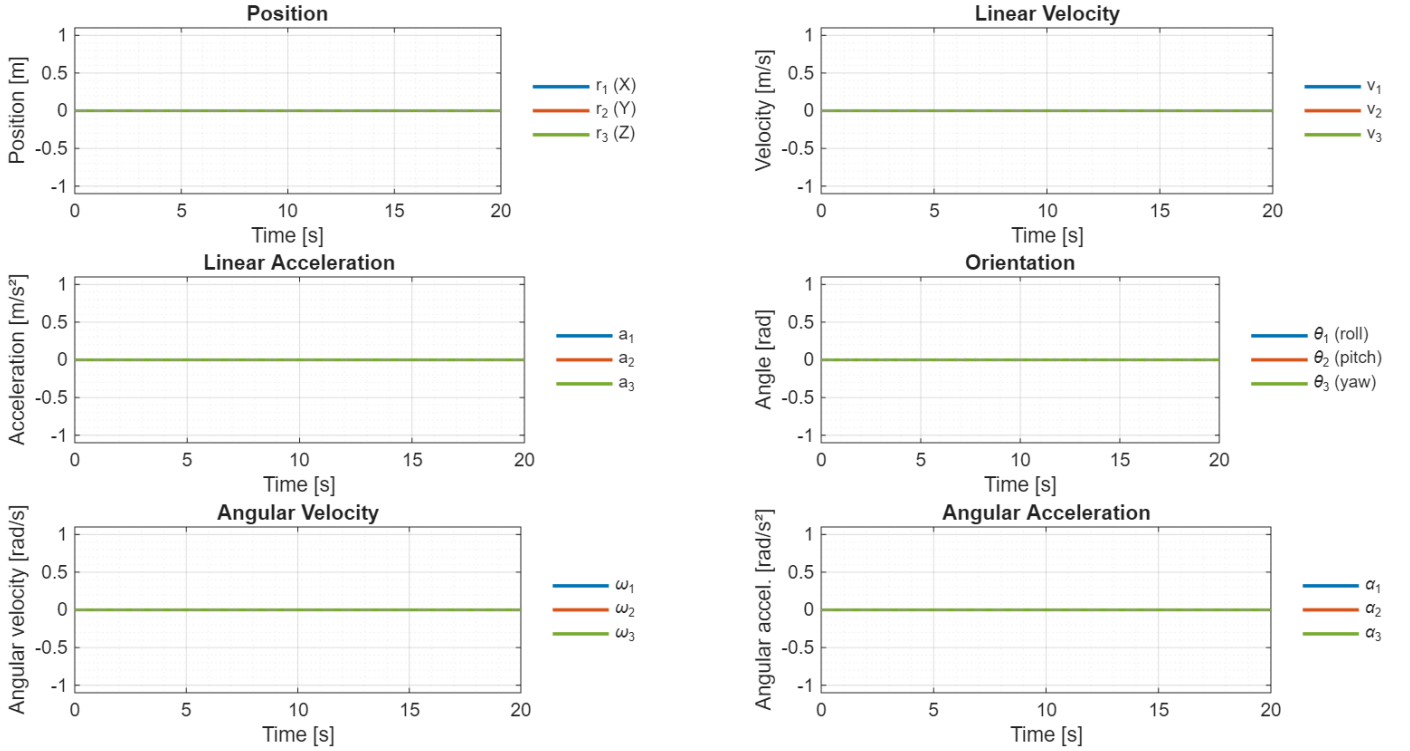


Figure 19: Q10 — Linearized model hover: all states remain at zero with zero perturbation input $F'_3 = 0$.

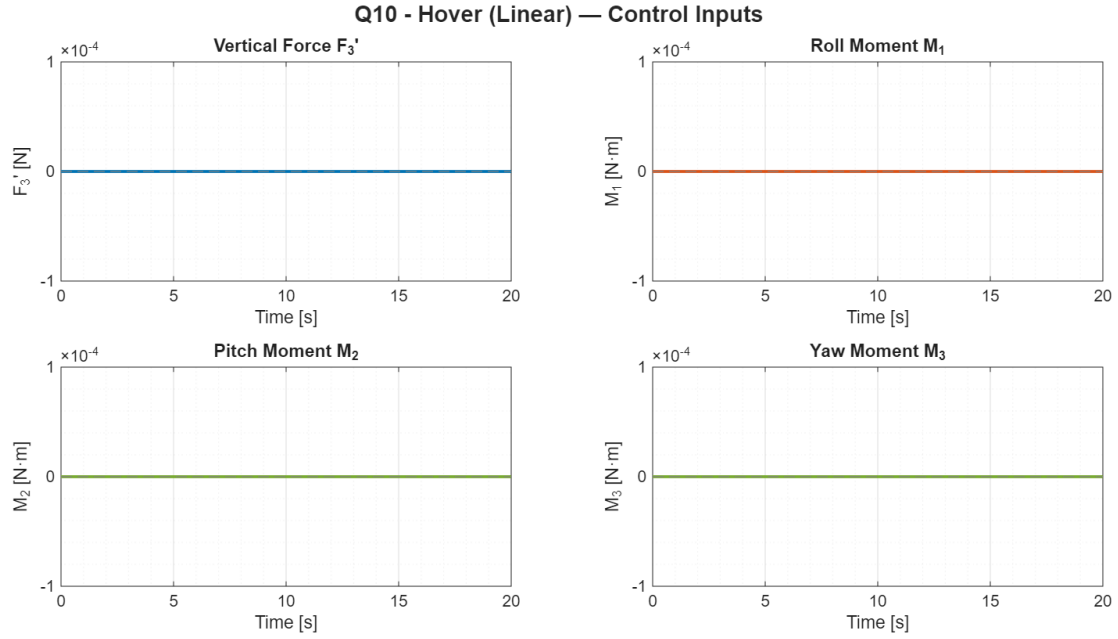


Figure 20: Q10 — Linearized model hover: zero perturbation input $F_3' = 0$ confirmed.

2.4 Comparison of Step Responses: Linear vs Nonlinear

The response of the linearized model is qualitatively consistent with the nonlinear model for the vertical force and yaw moment steps. The key difference appears in the pitch moment step response, where the nonlinear model exhibits oscillations due to Coriolis coupling between axes, while the linearized model — having dropped all quadratic velocity terms — responds smoothly with no cross-axis coupling.

2.4.1 Step in Vertical Force F_3'

A perturbation step F_3' is applied at $t = 2$ s. Since Subsystem 1 is a pure double integrator $H_1(s) = 1/(m_n s^2)$, r_3 grows parabolically and $\dot{r}_3$ ramps linearly, identical in character to the nonlinear response. All remaining states stay at zero, confirming full decoupling of the vertical subsystem. The linear and nonlinear models produce essentially identical responses here — since no rotation occurs, the small angle assumption holds exactly, the Coriolis terms are zero, and the rotation matrix remains $\mathbf{R}_n \approx \mathbf{I}$.

Q11₁ - Step in F3 (Linear) — States

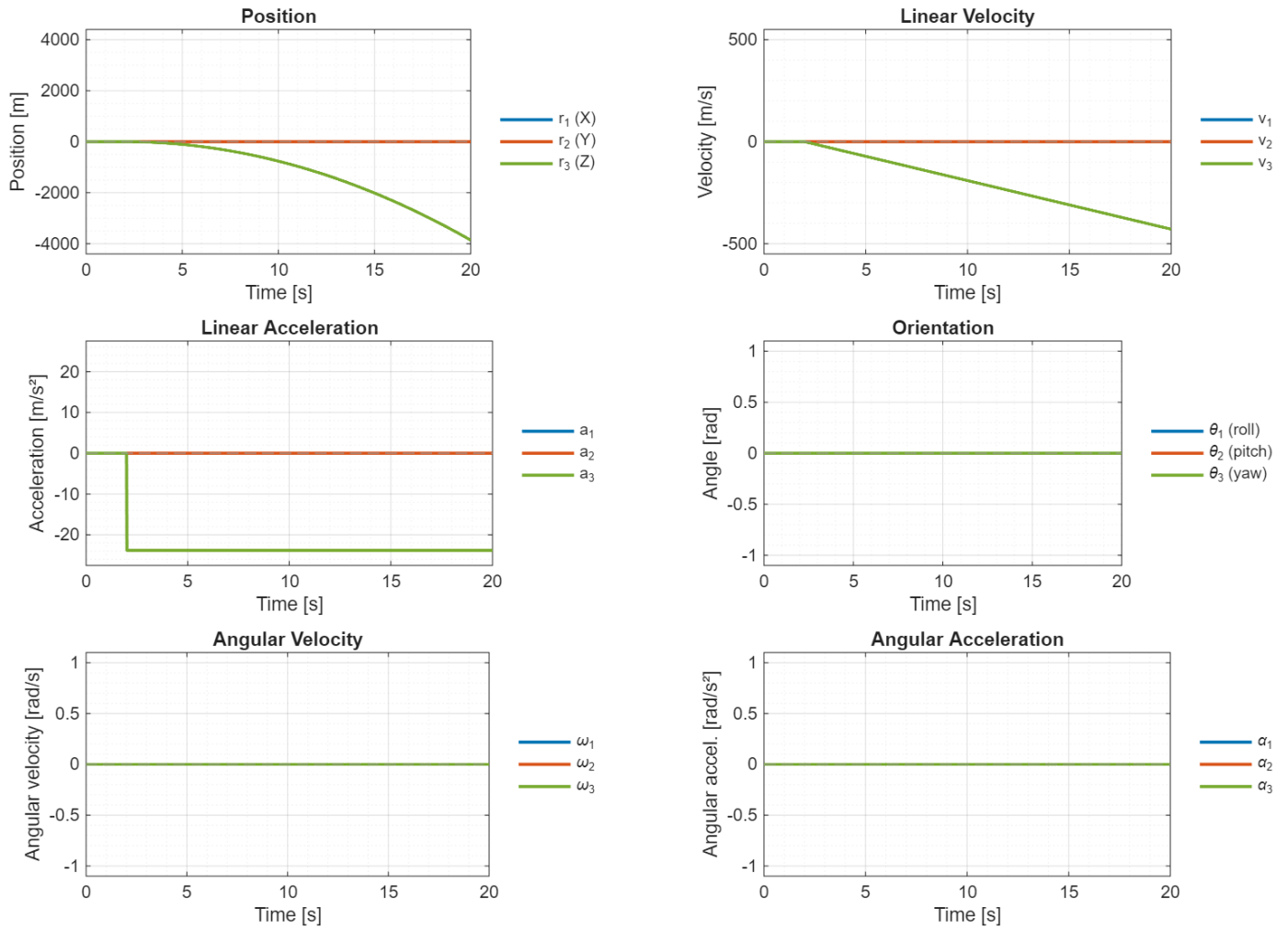


Figure 21: Q11 — Linearized model: states response to a step in F'_3 .

Q11₁ - Step in F₃ (Linear) — Control Inputs

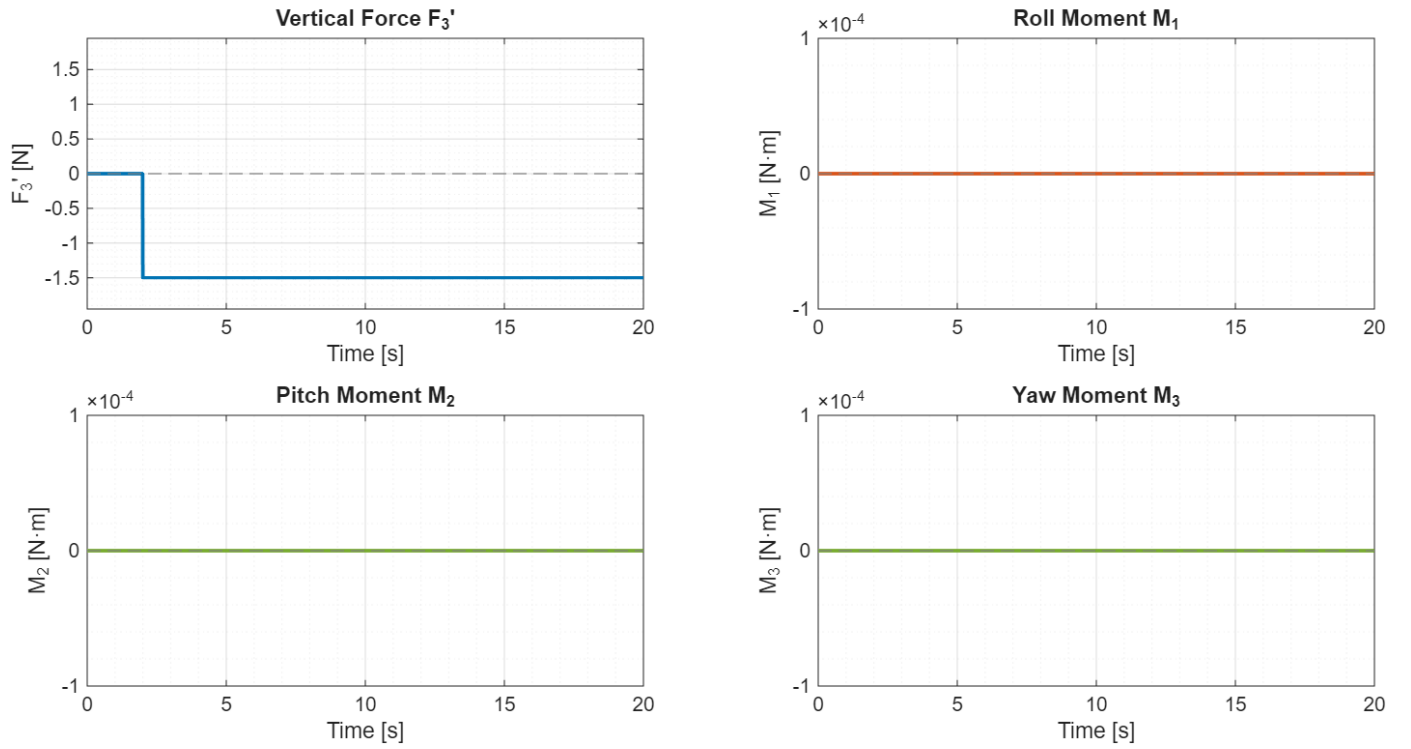


Figure 22: Q11 — Linearized model: input F_3' step signal.

2.4.2 Step in Yaw Moment M_3'

The linearized and nonlinear models produce essentially identical responses in this case. This is expected since yaw rotation does not induce any translational motion in either model, and the gyroscopic coupling terms in Equation 2.19 of the Quadcopter Dynamics guide remain zero as only ω_3 is nonzero. This is therefore one of the cases where the linearized model faithfully represents the true nonlinear dynamics.

Q11₂ - Step in M3 Yaw (Linear) — States

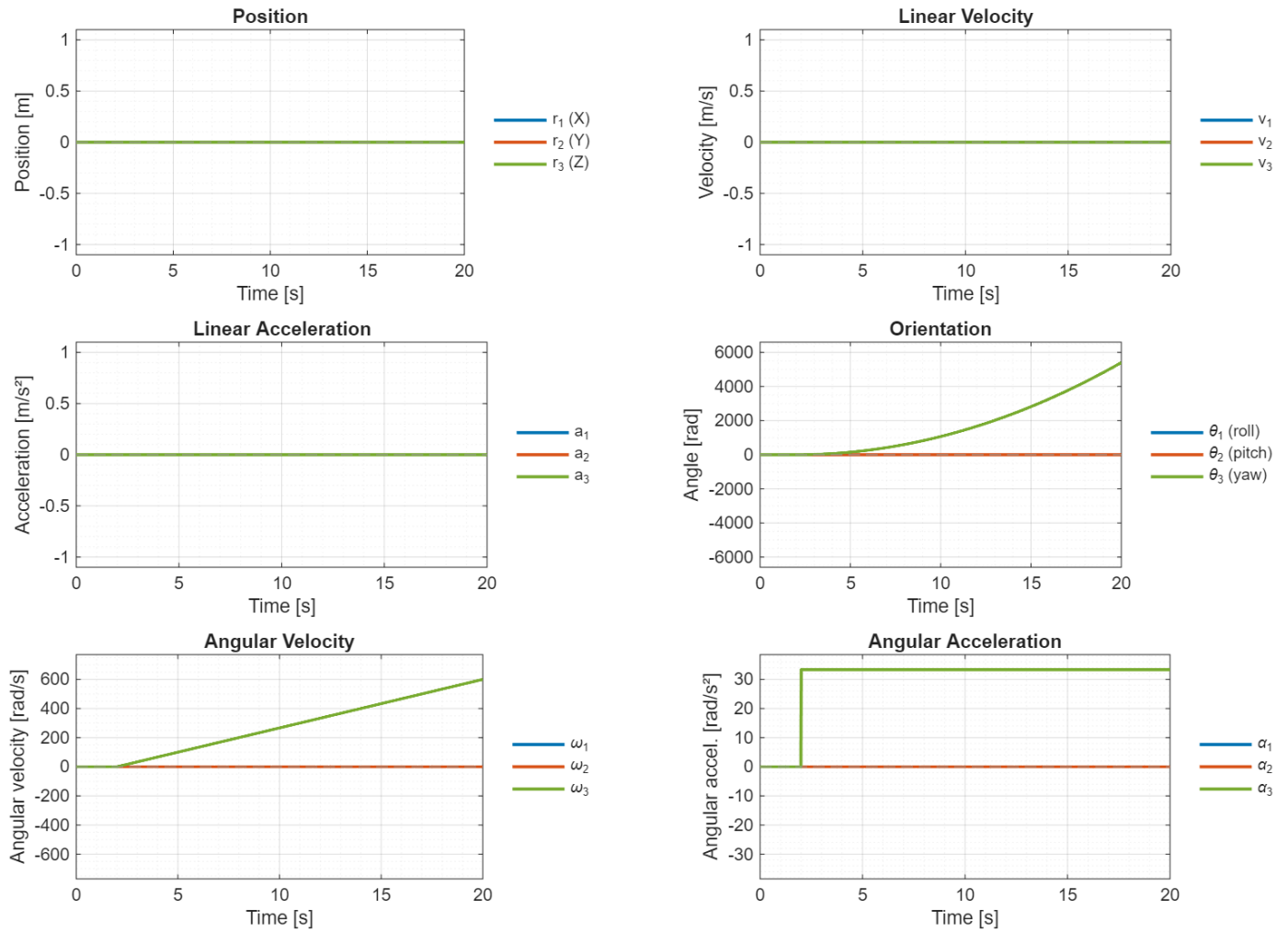


Figure 23: Q11 — Linearized model: states response to a step in M'_3 .

Q11₂ - Step in M3 Yaw (Linear) — Control Inputs

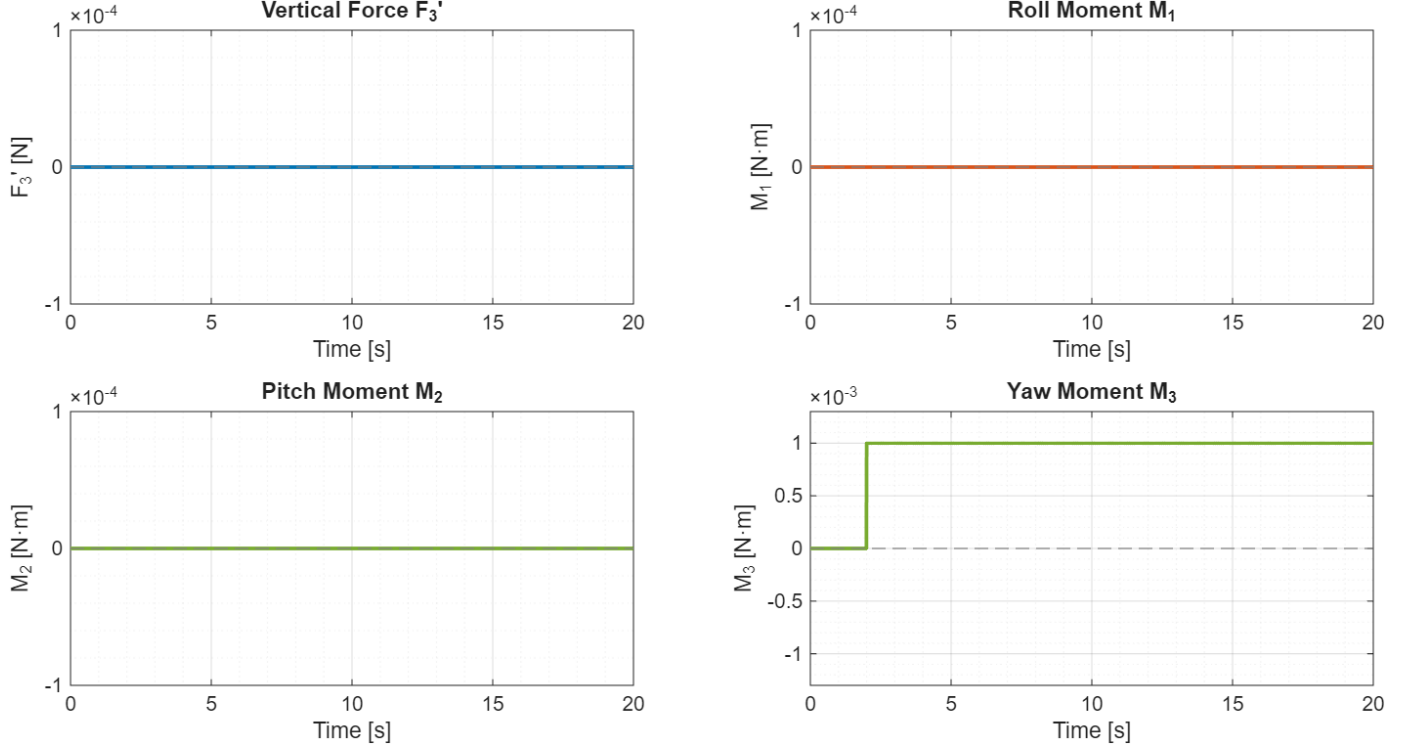


Figure 24: Q11 — Linearized model: input M_3' step signal.

2.4.3 Step in Pitch Moment M_2'

A perturbation step $M_2' = 5 \times 10^{-5}$ N·m is applied at $t = 2$ s while hovering. The expected pitch angular acceleration is:

$$\ddot{\theta}_2 = \frac{M_2'}{J_{22}} = \frac{5 \times 10^{-5}}{7.1691 \times 10^{-5}} \approx 0.698 \text{ rad/s}^2 \quad (28)$$

As θ_2 grows parabolically, gravity projects onto the longitudinal axis via $\ddot{r}_1 = -g\theta_2$, causing r_1 to drift in the negative X direction. All remaining states (r_2, θ_1, θ_3) stay at zero throughout.

Comparing with the nonlinear model, two important differences are observed. First, r_3 remains at zero in the linearized model — the small angle assumption ($\cos(\theta_2) \approx 1$) means the hover thrust continues to perfectly balance gravity regardless of pitch angle, whereas in the nonlinear model the tilting thrust vector causes r_3 to drift downward. Second, the oscillatory behaviour in the linear accelerations observed in the nonlinear model is completely absent here — without the Coriolis terms $v_i \times \omega_j$, the subsystems remain perfectly decoupled and the response is smooth. These two differences highlight where the linearized model diverges from reality as the drone moves away from the hover equilibrium.

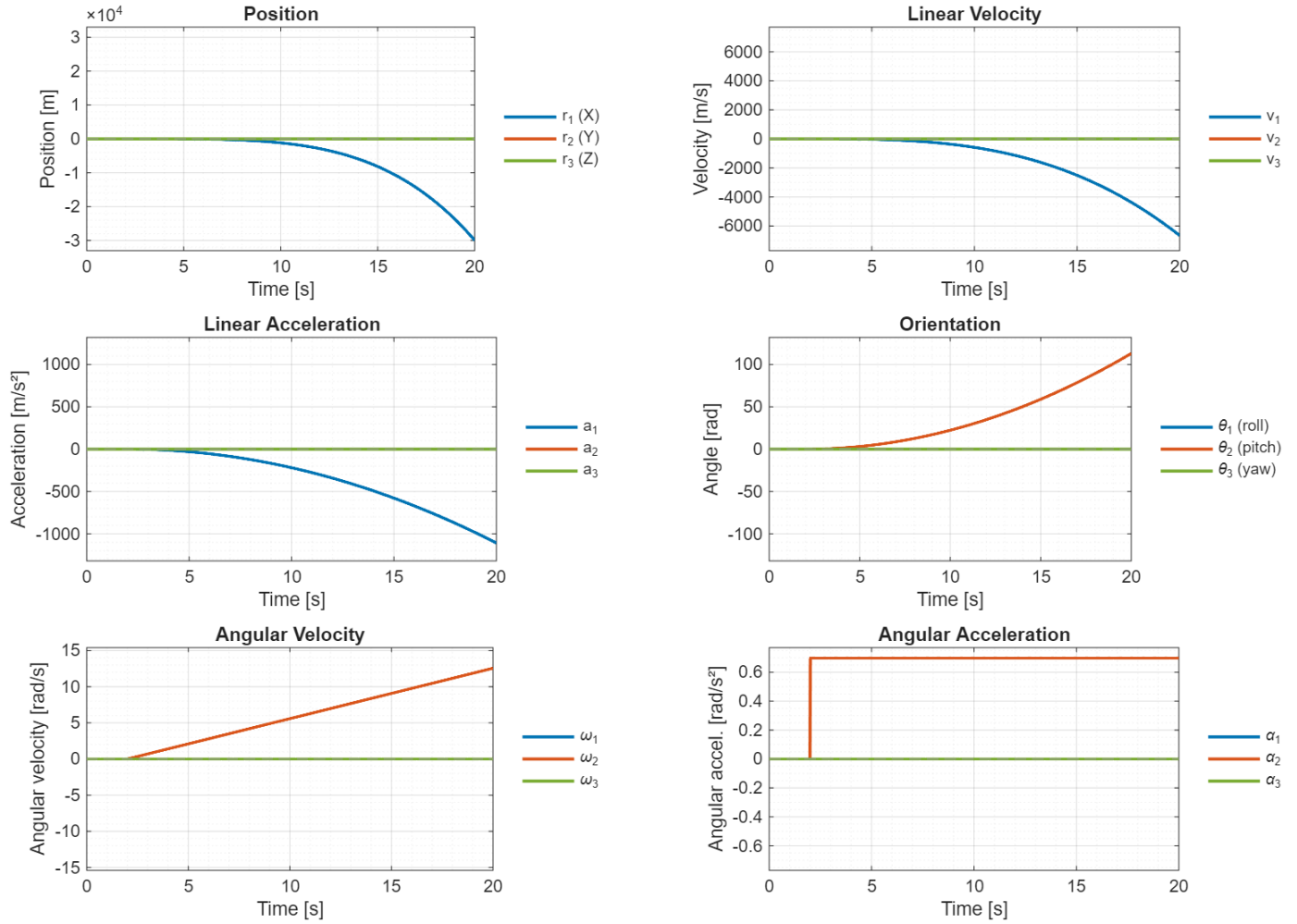
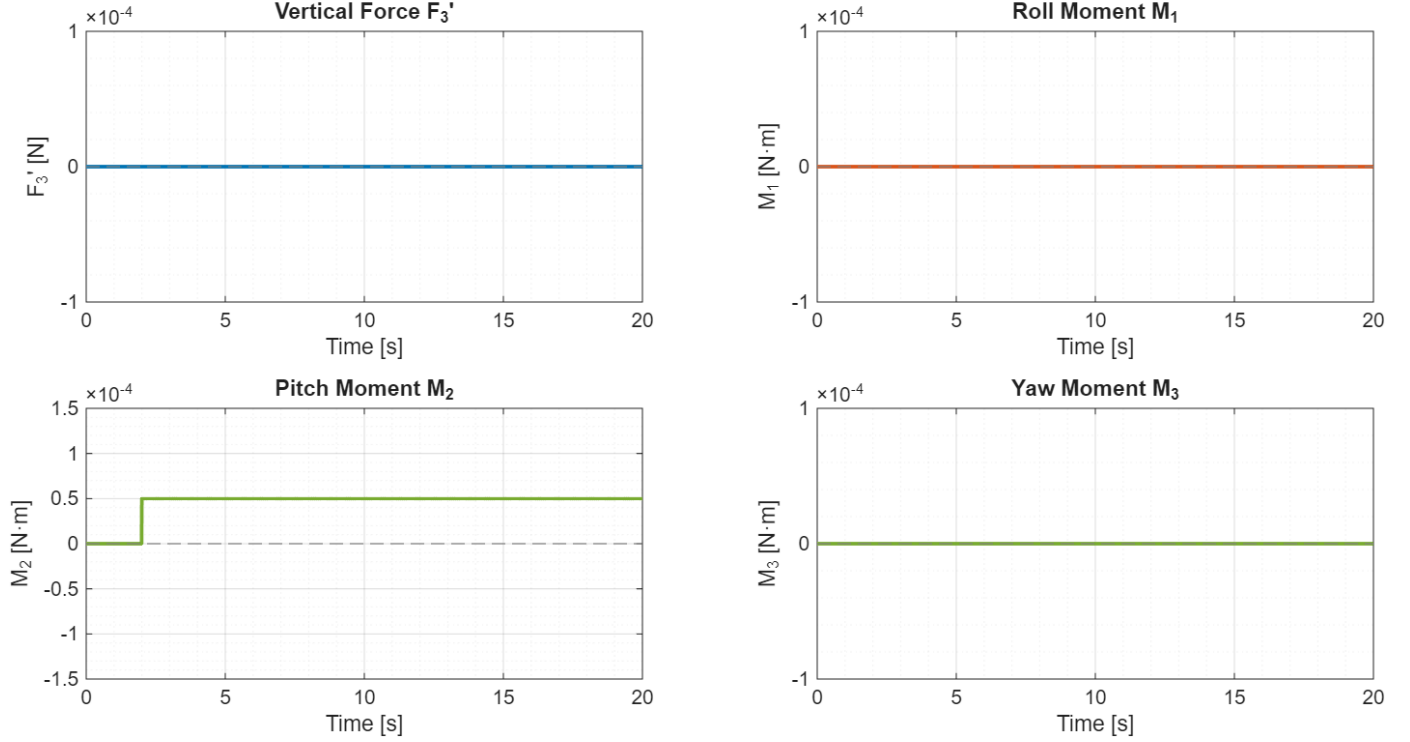
Q11₃ - Step in M2 Pitch (Linear) — States


Figure 25: Q11 — Linearized model: states response to a step in M'_2 . Unlike the nonlinear model, no oscillations are present as the Coriolis coupling terms are absent in the linearized equations.

Q11₃ - Step in M2 Pitch (Linear) — Control Inputs

Figure 26: Q11 — Linearized model: input M_2' step signal.

2.5 Takeoff, Rotate and Land on the Linearized Model

Both the angular and linear displacements with respect to the desired trajectory are the same as those obtained from the nonlinear model, confirming that the linearized model correctly captures the trajectory dynamics near the hover equilibrium. It is worth noting that the input vertical force differs from the one applied to the nonlinear system: in the linearized model it represents only the perturbation F_3' around the hover thrust, excluding the gravitational compensation term $-mg$. The moments are identical in both models, since $M_{eq} = 0$ at hover.

Q12 - Takeoff Rotate and Land (Linear) — States

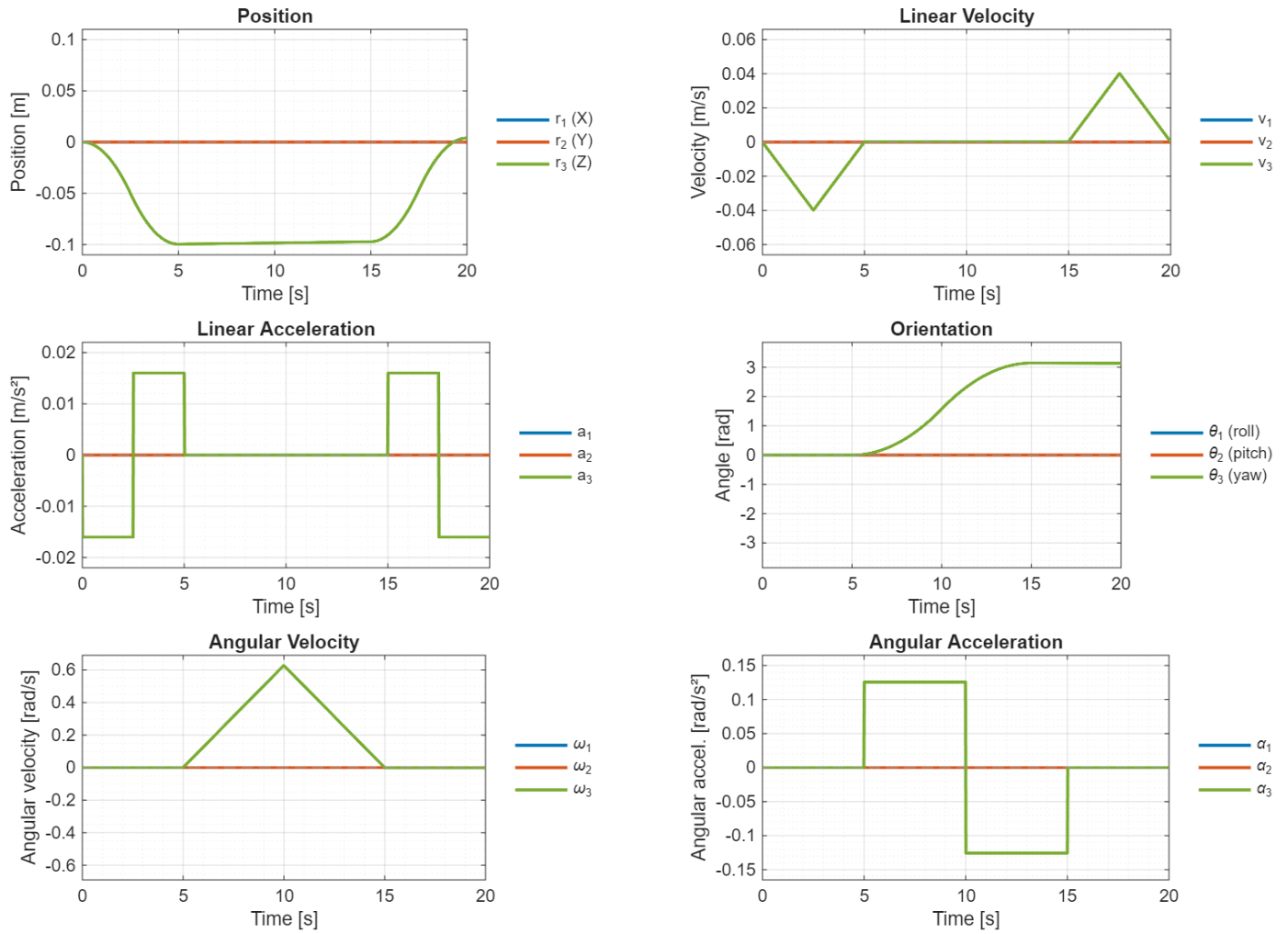
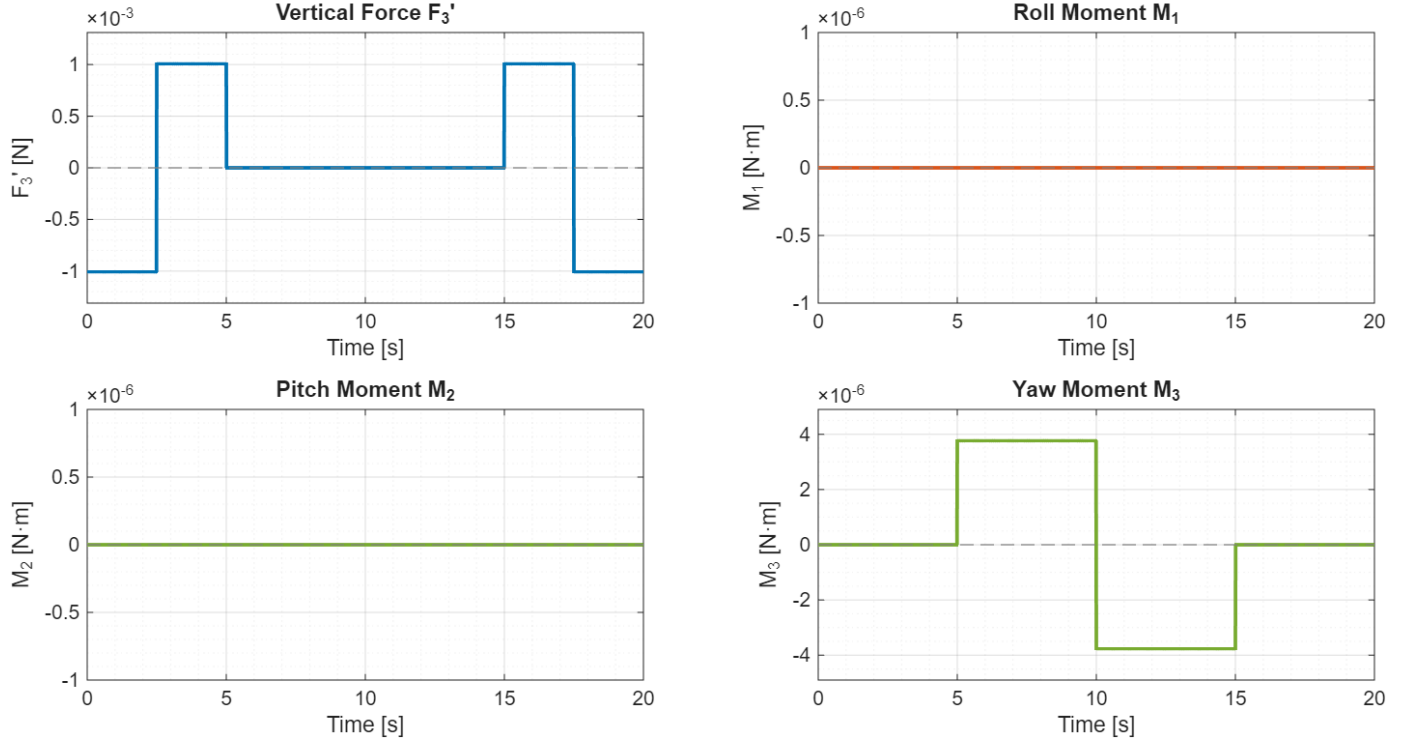


Figure 27: Q12 — Linearized model: position and orientation during takeoff, rotate and land routine.

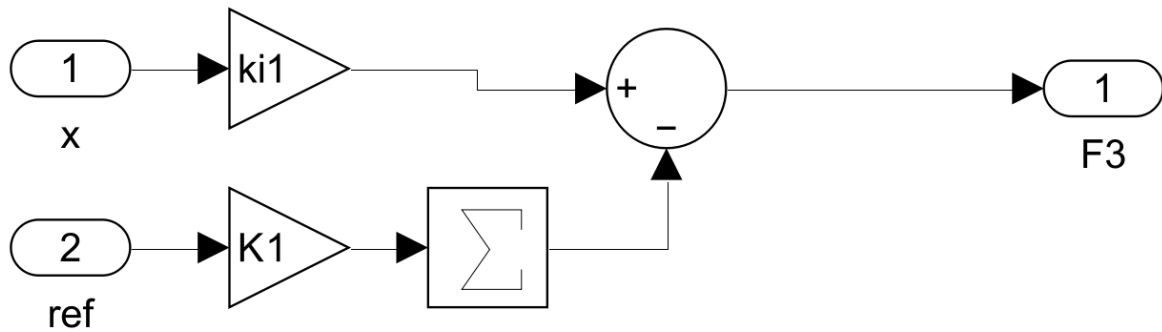
Q12 - Takeoff Rotate and Land (Linear) — Control Inputs

Figure 28: Q12 — Linearized model: control inputs F_3' and M_3' during the routine.

3 Full-State Feedback Regulator by Pole Placement

3.1 Role of the Regulator and Block Diagram

The regulator is the core component responsible for stabilizing the quadcopter and driving all states to the desired reference values. It implements a full-state feedback law of the form

$$\mathbf{u} = \mathbf{k}_i \mathbf{x} - K r_{\text{ref}}, \quad (29)$$


Figure 29: Regulator block used on both linear and non linear system

The quadcopter dynamics are decomposed into four decoupled subsystems, each governed by an independent regulator as in Figure 29

For each subsystem i , the gain vector $\mathbf{K}_i$ is computed using Ackermann's formula

$$\mathbf{K}_i = \mathbf{e}_n^\top \mathbf{C}_i^{-1} p(\mathbf{A}_i), \quad (30)$$

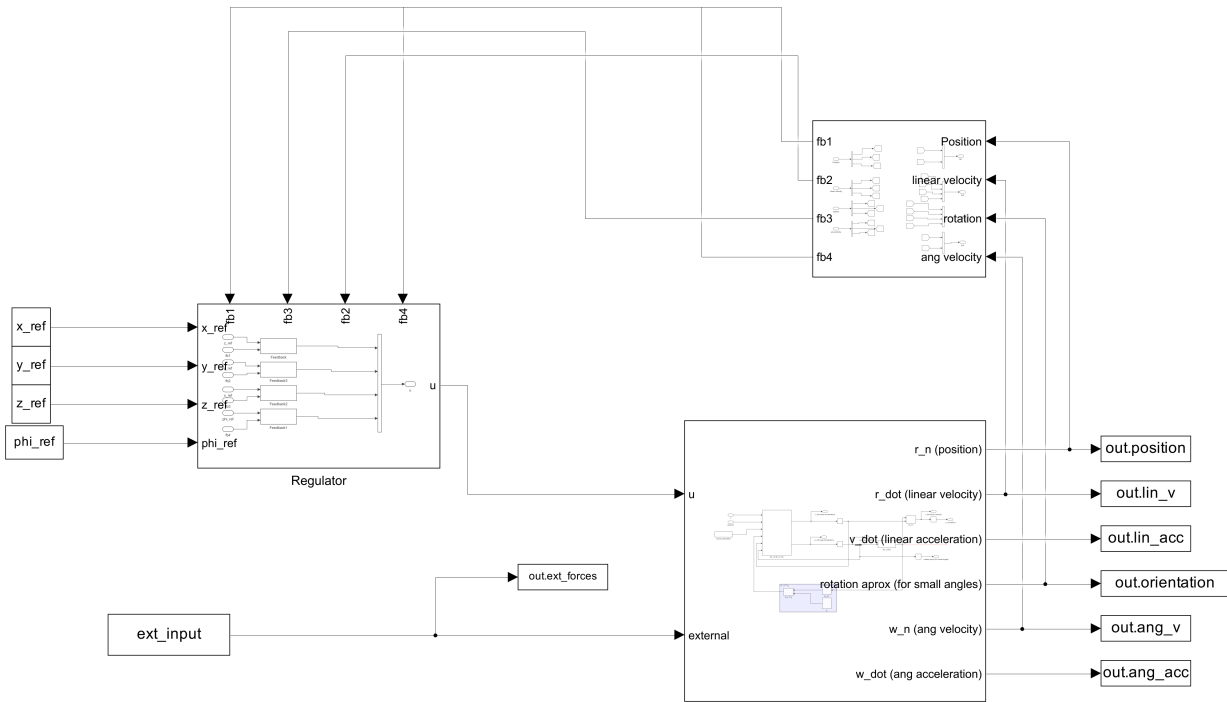


Figure 30: Non linear model from Simulink with regulator

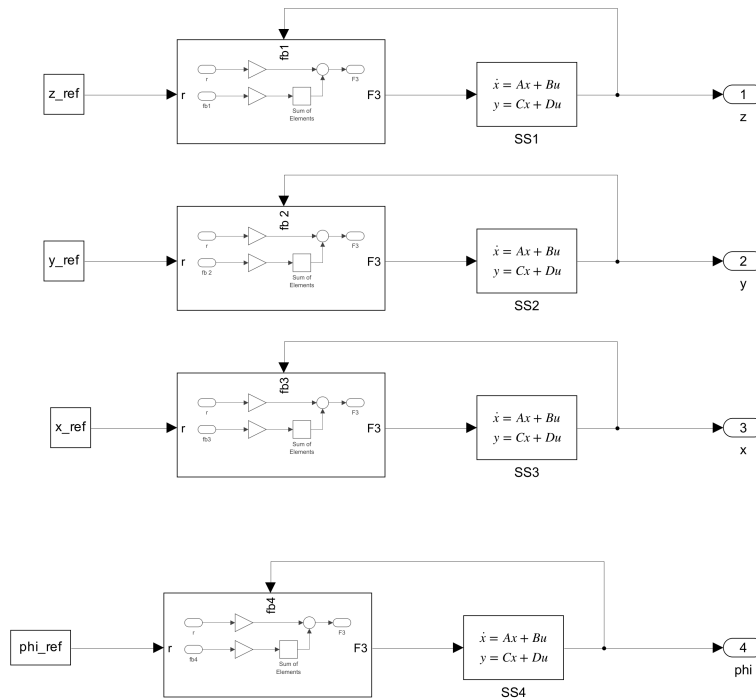


Figure 31: Linear model from Simulink with regulator

3.2 Convergence to Equilibrium with Single Subsystem Perturbation

All the poles used belong to the negative part of the real axis, ensuring closed-loop stability. The poles chosen for each subsystem are:

1. Vertical displacement subsystem (SS1 – F3): $[-3, -4]$

2. Lateral displacement subsystem (SS2 – M1): $[-4, -5, -6, -7]$
3. Longitudinal displacement subsystem (SS3 – M2): $[-3, -4, -5, -6]$
4. Yaw rotation angle subsystem (SS4 – M3): $[-2, -3]$

The reference setpoint used throughout this section is given in Table 2.

x [m]	y [m]	z [m]	ψ [rad]
2	1	4	0.5

Table 2: Reference values used for model validation.

For both simulations the initial condition was set to $r_3(0) = 0.5$ m, with all other states at zero.

Linear Model | IC: $r_3 = 0.5$ m (altitude)

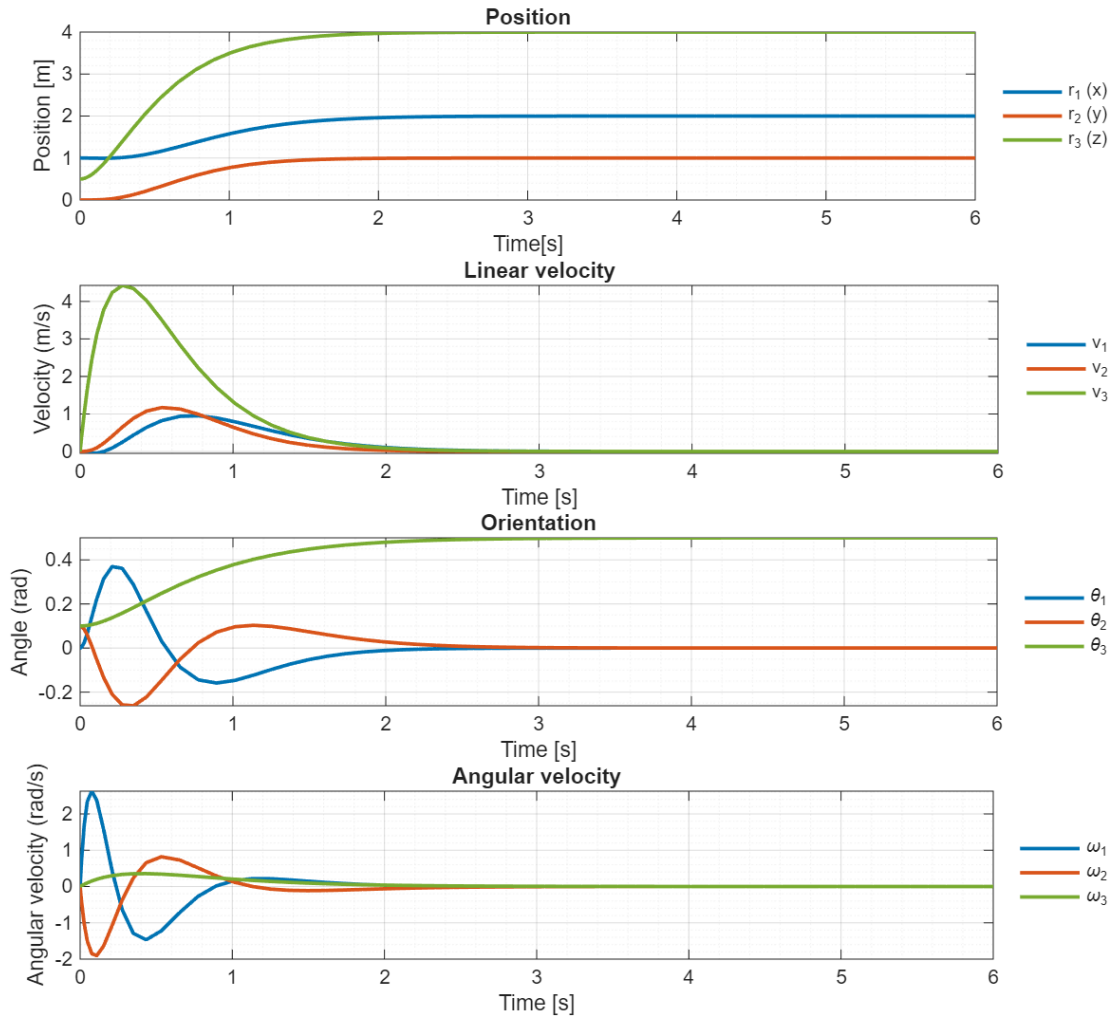


Figure 32: Linear model response for $r_3(0) = 0.5$ m: position, velocity, orientation, and angular velocity converge to the reference.

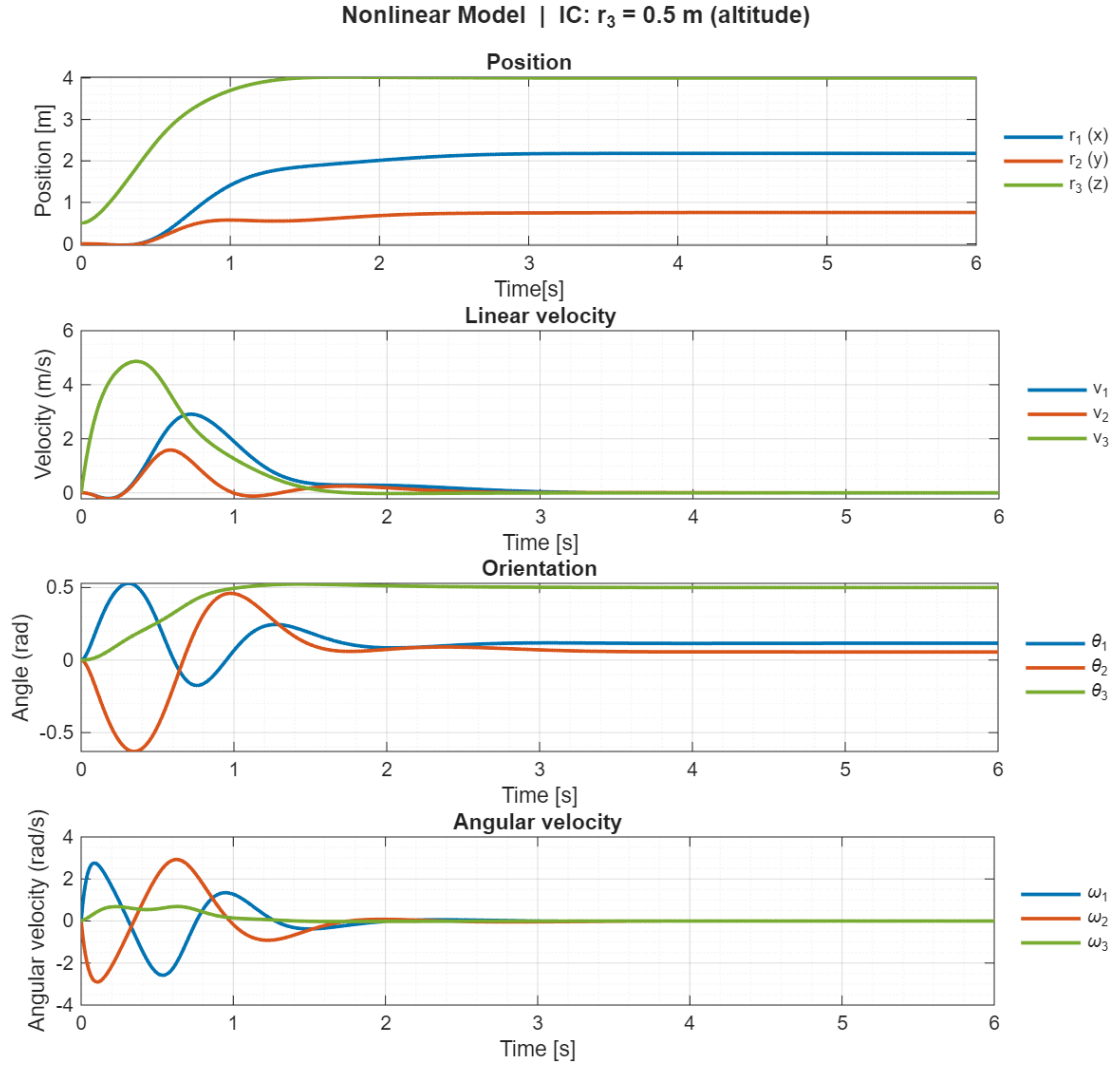


Figure 33: Nonlinear model response for the same initial condition and reference.

By analysis of Figure 32 and Figure 33, both the linear and nonlinear models converge to the equilibrium point defined by Table 2. The altitude state r_3 rises from its initial offset and settles at $z = 4$ m. The remaining states, which start at zero, are driven to their respective reference values with no steady-state error. This confirms that the pole-placement and gains computed via Ackermann's formula are valid for both model representations.

3.3 Linear vs. Nonlinear: Temporal and Control Action Comparison

To assess the validity of the linearisation, the responses of both models were overlaid for the same initial condition ($r_3(0) = 0.5$ m) and the reference from Table 2.

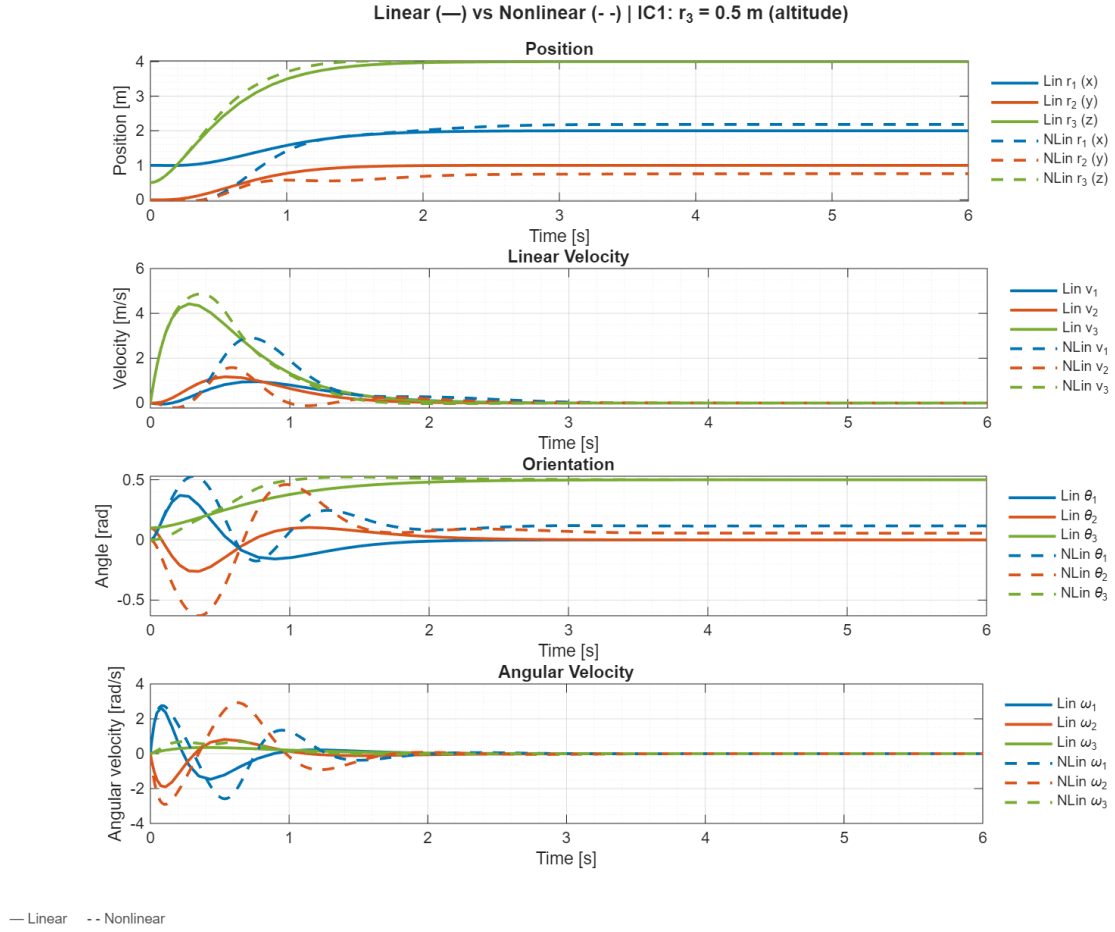


Figure 34: Linear (solid) vs. Nonlinear (dashed) response for $r_3(0) = 0.5$ m.

The two models show close agreement in the altitude channel: both converge to $z = 4$ m with similar settling times and negligible steady-state error. Small discrepancies are visible in the transient of the coupled lateral and pitch states, which rise because the nonlinear model retains the coupling terms that the linear model discards. Overall, for this small initial perturbation the linear model is a reliable approximation of the true nonlinear dynamics.

3.4 Multi-Subsystem Simultaneous Initial Condition Analysis

To verify robustness of the regulator when multiple subsystems are simultaneously perturbed, the initial conditions were set to $r_3 = 0.5$ m, $r_1 = 1.0$ m, $v_2 = 2$ m/s, $\theta_1 = 0.1$ rad, $\theta_2 = 0.1$ rad, and $\omega_1 = 0.2$ rad/s, while keeping the same reference values and poles described in subsection 3.2.

Linear Model | IC: $r_3=0.5$ m, $r_1=1$ m, $v_2 = 2$ m/s, $\theta_1=0.1$ rad, $\theta_2=0.1$ rad, $\theta_3= 0.1$ w1 = 0.2 rad/s

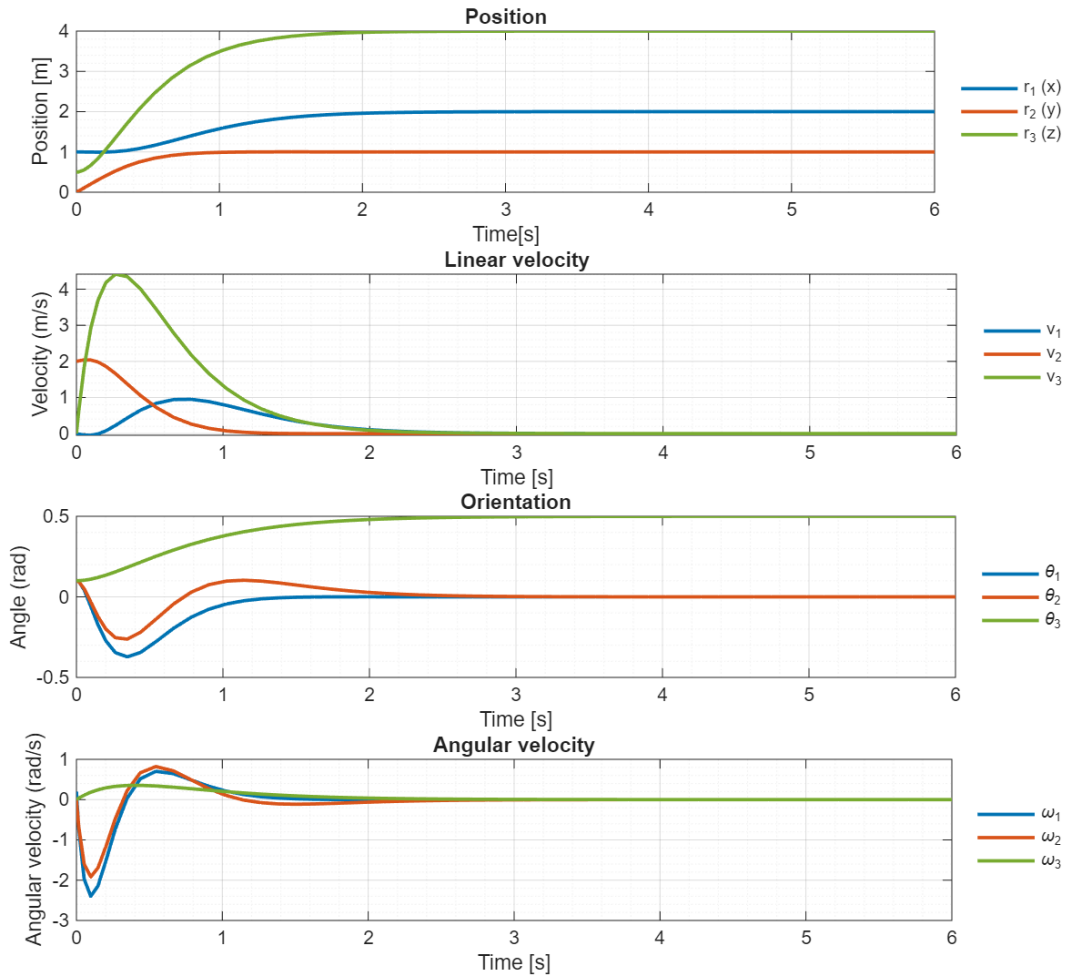


Figure 35: Linear model response under multi-subsystem initial conditions.

Nonlinear Model | IC: $r_3=0.5$ m, $r_1=1$ m, $v_2 = 2$ m/s, $\theta_1=0.1$ rad, $\theta_2=0.1$ rad, $\theta_3=0.1$ w1 = 0.2 rad/s

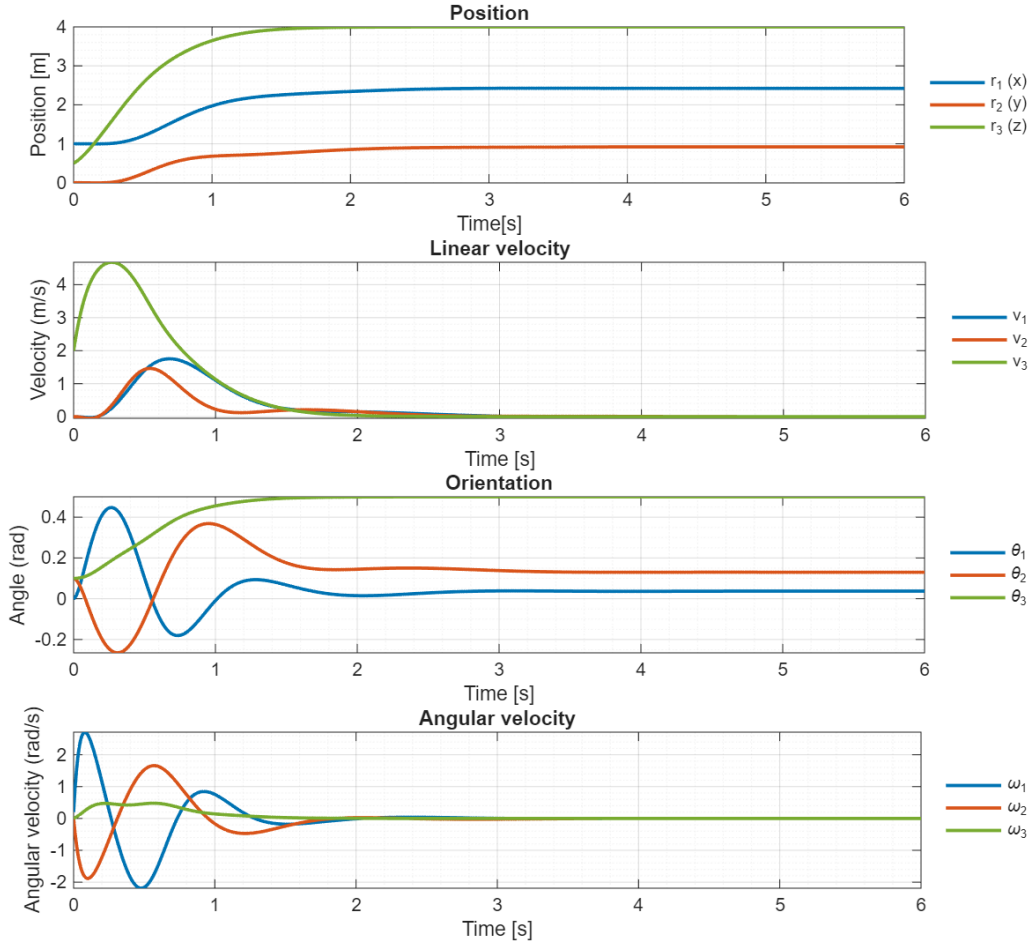


Figure 36: Nonlinear model response under multi-subsystem initial conditions.

Both models converge to the equilibrium despite the simultaneous perturbations across all four subsystems. The linear model settles smoothly, consistent with the decoupled dynamics assumed by the regulator design. The nonlinear model exhibits slightly larger transient oscillations in the lateral and pitch channels due to cross-coupling, but still converges to the reference. This confirms that the full-state feedback regulator is robust to multi-axis perturbations of moderate magnitude.

3.5 Effect of Dominant Complex Poles on Closed-Loop Performance

To investigate the effect of pole location on the transient response, the poles were redesigned using a dominant complex conjugate pair with damping ratio $\zeta = 0.7$ and natural frequency $\omega_n = 3$ rad/s. This design targets a settling time of approximately 1.4 s and an overshoot of around 5 %. For the fourth-order subsystems SS2 and SS3, two additional fast real poles are placed four times further left than the dominant pair, making their transient contribution negligible. With $\sigma = \zeta \omega_n = 2.1$ and $\omega_d = \omega_n \sqrt{1 - \zeta^2} \approx 2.14$ rad/s, the new poles are listed in Table 3.

Subsystem	Description	Order	Poles
SS1	Vertical displacement (F3)	2	$-2.1 \pm 2.14j$
SS2	Lateral displacement (M1)	4	$-2.1 \pm 2.14j, -8.4, -9.4$
SS3	Longitudinal displacement (M2)	4	$-2.1 \pm 2.14j, -8.4, -9.4$
SS4	Yaw rotation (M3)	2	$-2.0, -2.5$

Table 3: Dominant complex poles used for Q17–Q18 simulations.

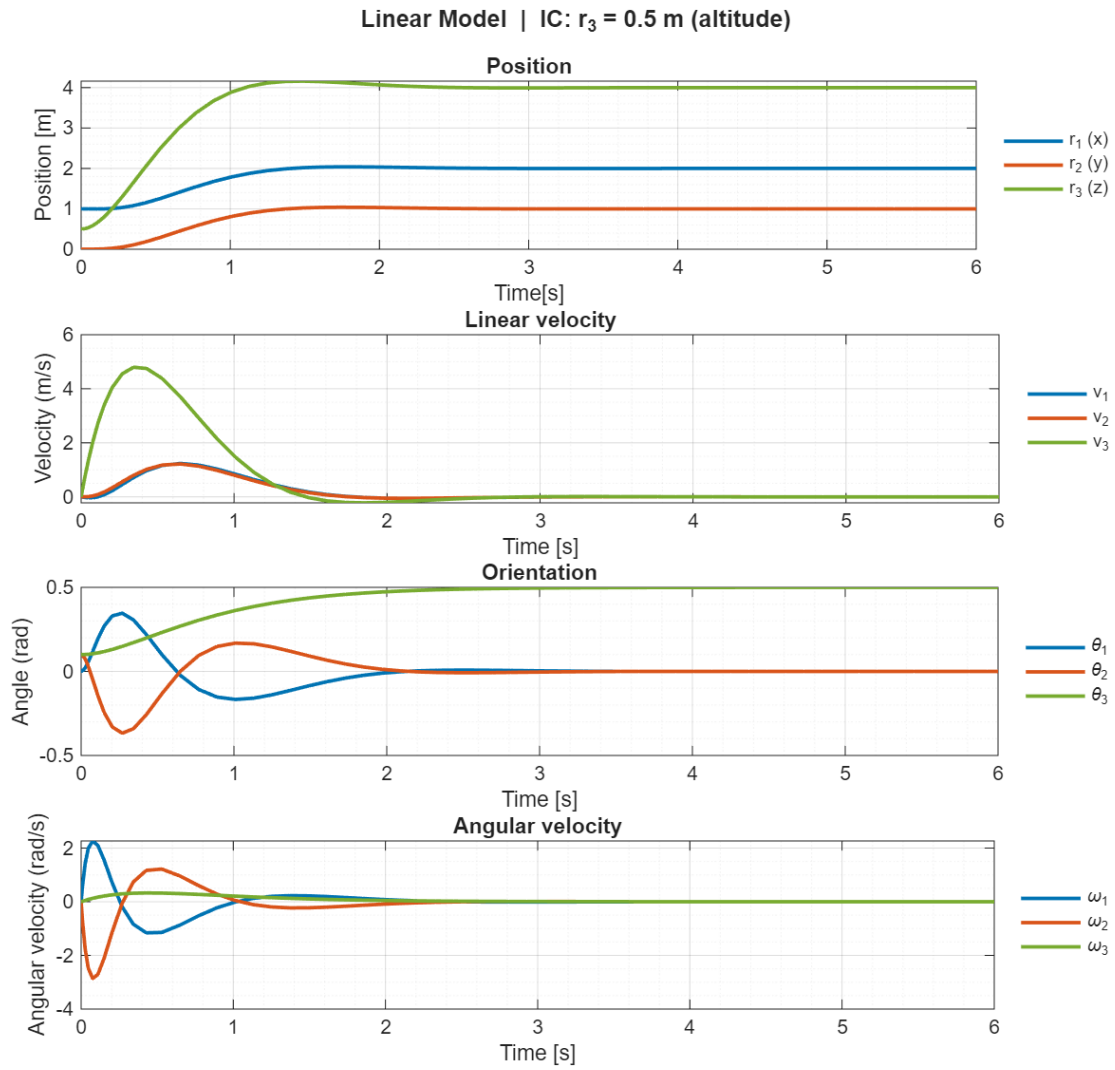


Figure 37: Linear model response with dominant complex poles, $r_3(0) = 0.5$ m.

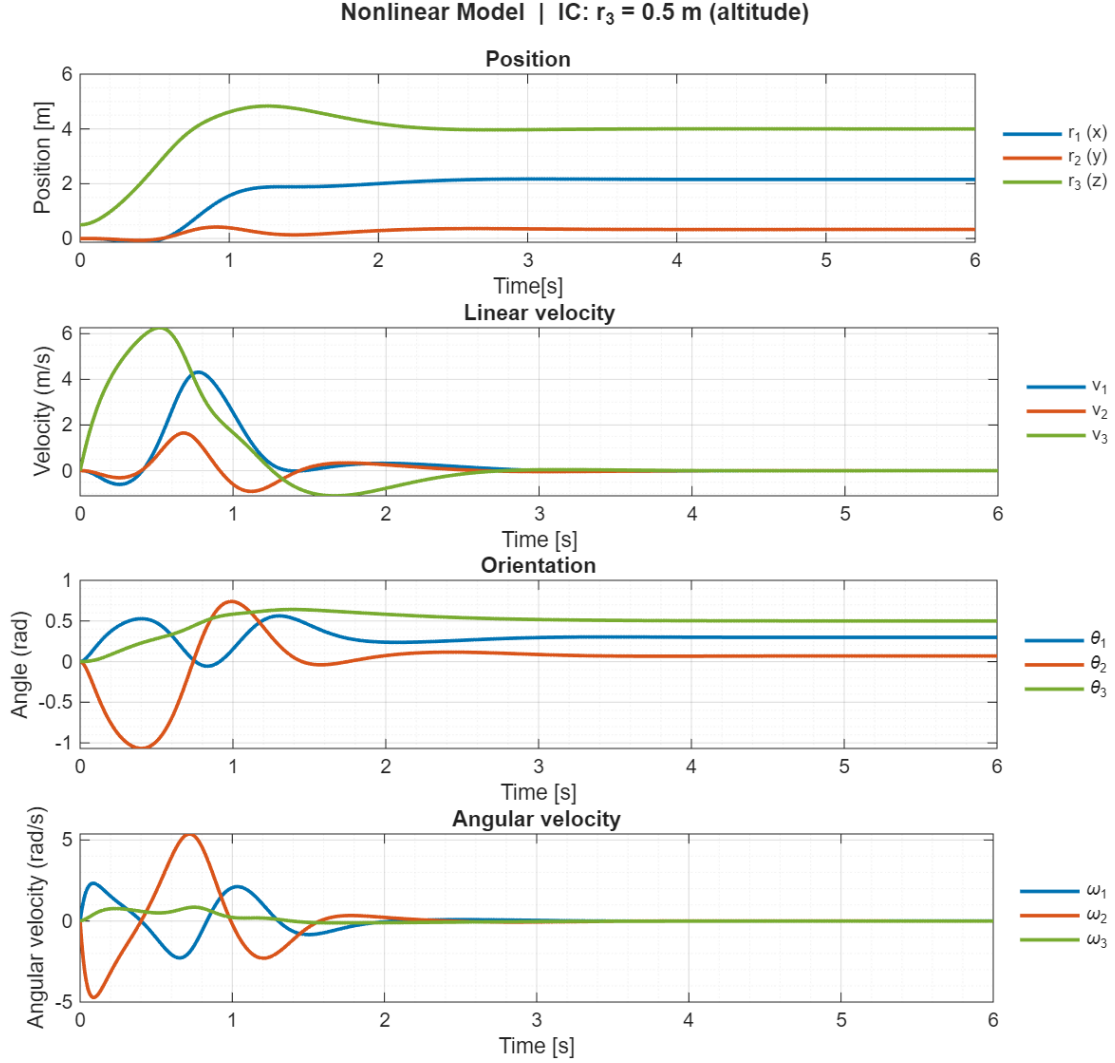


Figure 38: Nonlinear model response with dominant complex poles, $r_3(0) = 0.5$ m.

Compared to the real-pole results in subsection 3.2, the complex dominant poles introduce a small overshoot ($\approx 5\%$) but achieve a noticeably faster settling time. Both the linear and nonlinear models converge to equilibrium, with the nonlinear model showing slightly more pronounced transients due to the coupling terms absent from the linearisation.

3.6 Influence of Damping Ratio and Natural Frequency on Closed-Loop Behavior

Building on the dominant-pair design of subsection 3.5, the damping ratio ζ and natural frequency ω_n were systematically varied for the vertical (SS1) and longitudinal (SS3) subsystems, while SS2 and SS4 retained the poles from Table 3. Three values of damping $\zeta \in \{0.3, 0.7, 1.0\}$ and two natural frequencies $\omega_n \in \{1.5, 3.0\}$ rad/s were tested, getting the six configurations in Table 4.

Case	ζ	ω_n [rad/s]	Dominant poles $p_{1,2}$
1	0.3	1.5	$-0.45 \pm 1.43j$
2	0.3	3.0	$-0.90 \pm 2.86j$
3	0.7	1.5	$-1.05 \pm 1.07j$
4	0.7	3.0	$-2.10 \pm 2.14j$ (Q17 design)
5	1.0	1.5	$-1.5, -1.5$
6	1.0	3.0	$-3.0, -3.0$

Table 4: Pole configurations tested for SS1 and SS3 in Q18. Case 4 corresponds to the Q17 baseline.

The analysis is performed with initial condition $r_1(0) = 1$ m (longitudinal offset), chosen to directly excite SS3 and observe the effect of each pole configuration on the longitudinal and pitch dynamics. All six cases are overlaid on the same figure for direct comparison between linear and nonlinear models.

Position and Velocity

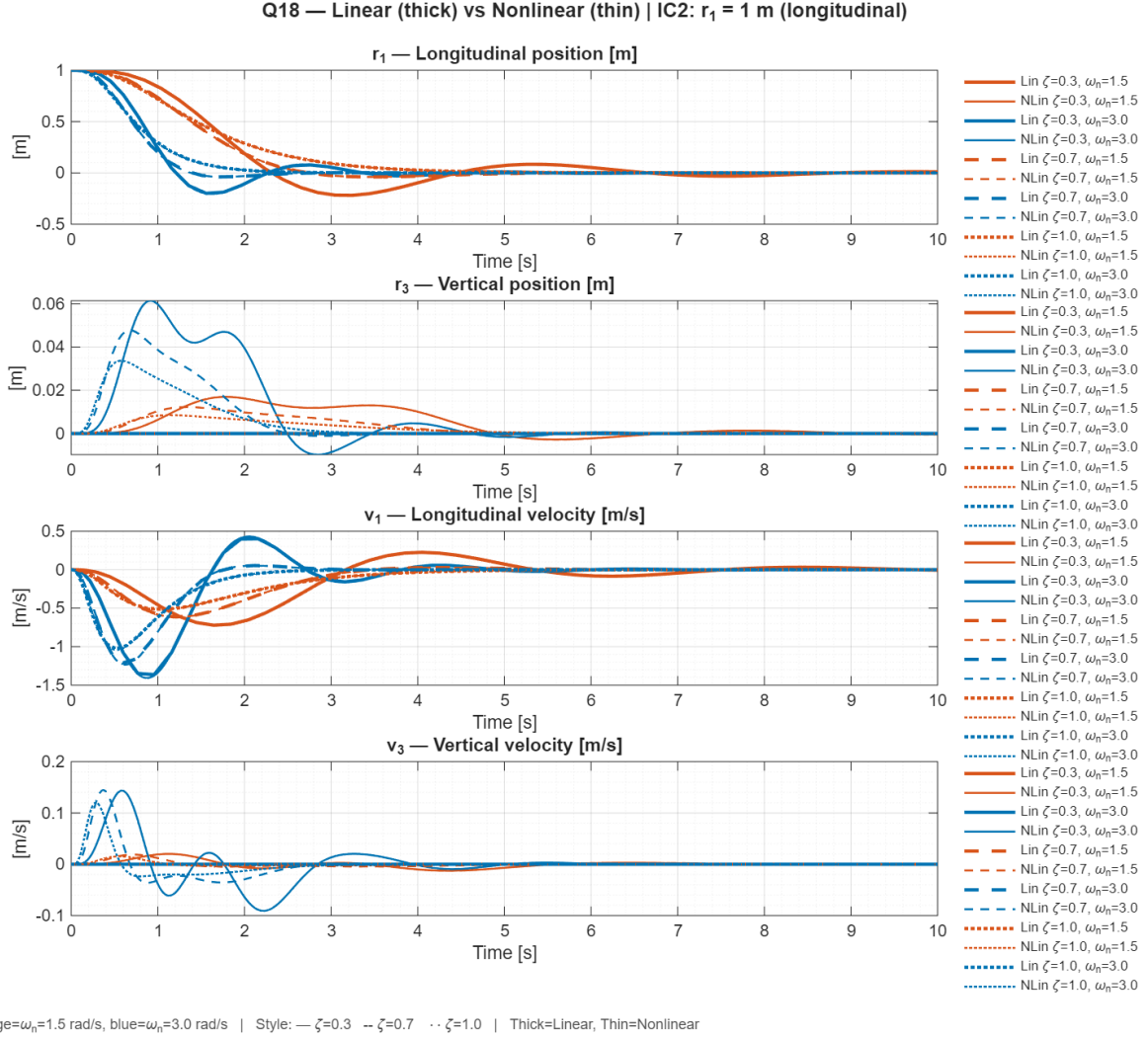


Figure 39: Linear (thick) vs. Nonlinear (thin): longitudinal position r_1 , vertical position r_3 , and corresponding velocities v_1 , v_3 for all six pole configurations. Initial condition $r_1(0) = 1$ m. Color encodes ω_n (orange = 1.5 rad/s, blue = 3.0 rad/s); line style encodes ζ (solid = 0.3, dashed = 0.7, dotted = 1.0).

The following observations can be made from Figure 39:

- **Effect of ζ :** The under damped cases ($\zeta = 0.3$) produce noticeable oscillations in r_1 and v_1 before settling. As ζ increases to 0.7 and 1.0 the overshoot is progressively reduced until it is entirely eliminated.
- **Effect of ω_n :** For a fixed ζ , increasing ω_n from 1.5 to 3.0 rad/s accelerates the response time.
- **Linear vs. Nonlinear:** The linear and nonlinear responses are very close alike for the well-damped cases ($\zeta = 0.7$ and $\zeta = 1.0$). For the under damped case ($\zeta = 0.3$), the nonlinear model exhibits slightly larger oscillation amplitudes and a slight increase on the settling time.

Pitch Orientation and Angular Velocity

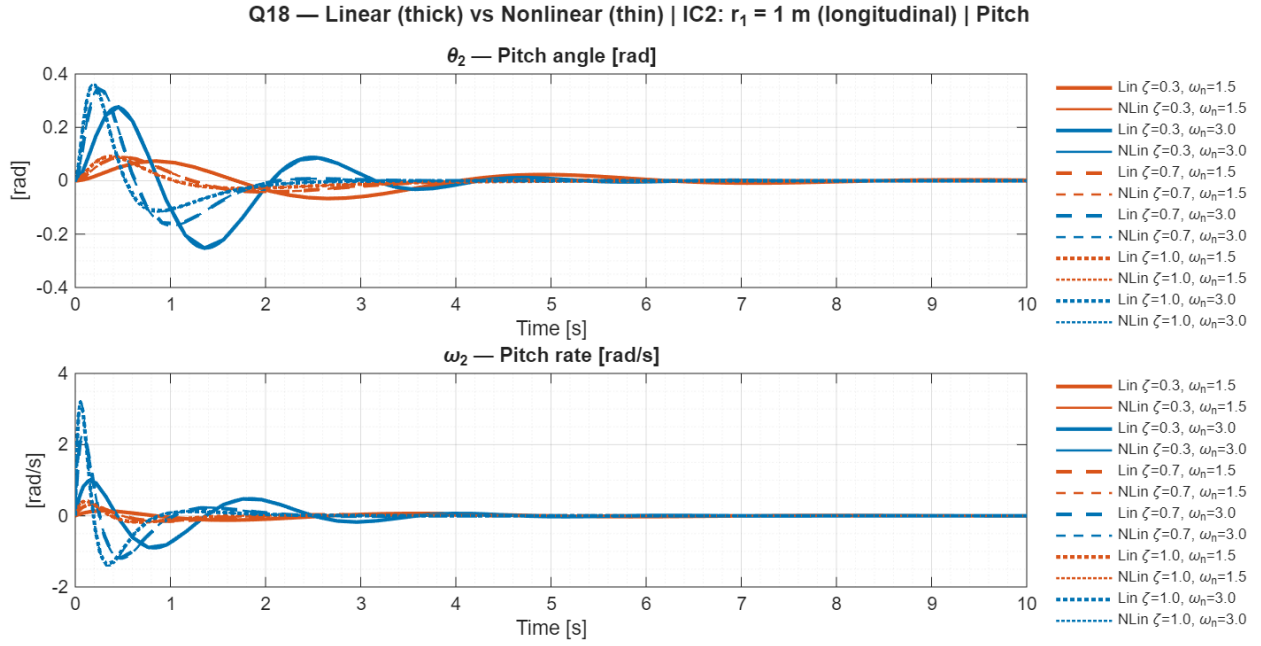


Figure 40: Linear (thick) vs. Nonlinear (thin): pitch angle θ_2 and pitch rate ω_2 for all six pole configurations. Initial condition $r_1(0) = 1$ m. Same color and line-style convention as Figure 39.

From Figure 40:

- **Pitch excitation:** A longitudinal position offset requires the regulator to generate a pitch moment to accelerate the drone towards the reference. The magnitude of the initial pitch angle θ_2 therefore increases with ω_n , since a faster desired response demands a larger correction.
- **Damping effect on pitch:** The under damped cases produce oscillatory pitch that mirrors the oscillations in r_1 . Critical damping eliminates the oscillations, producing a smooth return to zero pitch.
- **Linear vs. Nonlinear discrepancy:** The pitch channel shows the largest deviation between the two models. In the nonlinear model, the coupling between altitude and translational dynamics causes θ_2 to exhibit a slightly different transient shape compared to the linear prediction, particularly for $\zeta = 0.3$ and high ω_n .

3.7 Influence of Non-Dominant Poles on Longitudinal Dynamics

Q19 — Non-dominant pole influence on longitudinal dynamics
 $r_1(0) = 1$ m (longitudinal offset) | Linear (—) vs Nonlinear (- -)

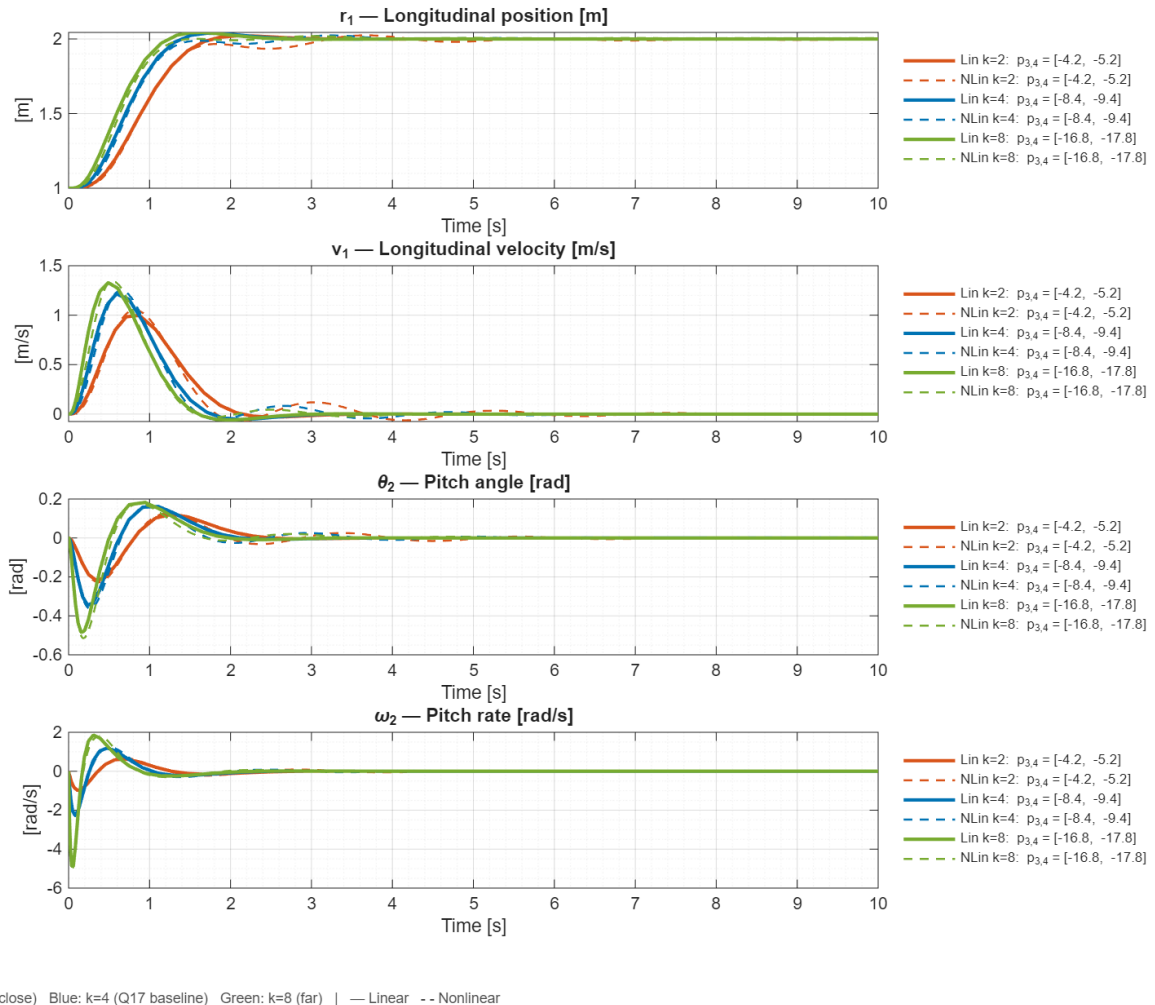


Figure 41: Q19 — Effect of non-dominant pole location on longitudinal dynamics. Linear (solid, thick) vs. Nonlinear (dashed, thin). Orange: $k = 2$ (close), Blue: $k = 4$ (Q17 baseline), Green: $k = 8$ (far). Dominant pair fixed at $-2.1 \pm 2.14j$ for all cases.

The following observations can be drawn from Figure 41:

- **Case 1 ($k = 2$, close non-dominant poles):** The non-dominant poles at -4.2 and -5.2 are twice as far from the origin like the dominant pair. Their contribution is non-negligible: the initial transient in r_1 and v_1 shows a faster decay imposed on the dominant oscillation, distorting the expected second-order shape. The pitch angle θ_2 also shows a sharper initial peak compared to the other cases. This confirms that non-dominant poles placed too close to the dominant pair cross the separation assumption and change the response.
- **Case 2 ($k = 4$, Q17 baseline):** With non-dominant poles at -8.4 and -9.4 , the response closely follows the ideal dominant-pair second-order behavior. The initial distortion visible in Case 1 is no longer present, and the curves for r_1 , v_1 , θ_2 , and ω_2 are smooth and consistent with the expected response. This shows the choice made in the Q17 design, was the correct one.
- **Case 3 ($k = 8$, far non-dominant poles):** The response is virtually indistinguishable from Case 2, confirming that beyond a ratio of $4\times$ the distance of the dominant poles from the origin, moving the non-dominant poles further has no practical effect on the response.
- **Linear vs. Nonlinear:** Across all three cases, the linear and nonlinear models are very similar. The largest

deviation occurs in Case 1, where the faster non-dominant dynamics interact with the nonlinear coupling between pitch and longitudinal motion, producing a slightly larger initial pitch angle in the nonlinear model.

4 Motor Saturation

4.1 Effect of Input Saturation on Closed-Loop Performance

Motor saturation arises when the demanded input surpasses the physical output capacity of the motors, clamping each command signal u_i within the admissible range $[0, u_{i,\max}]$. When the saturation block connected converting generalized forces to motor commands via Equation 2.16, applying the clamp, and converting back via Equation 2.15 the closed-loop system retains its ability to converge to the desired equilibrium, provided the pole placement does not demand excessive control effort.

To deliberately induce saturation, one non-dominant pole of the longitudinal subsystem was relocated to the right half of the s -plane. This renders the associated gain large enough that the required torque τ_y exceeds the actuators capacity. As a consequence, the effective generalized forces fed to the drone dynamics are permanently clamped at their saturation ceiling, depriving the controller of the capacity to stabilize. The mismatch between the demanded and delivered forces causes the system to lose stability.

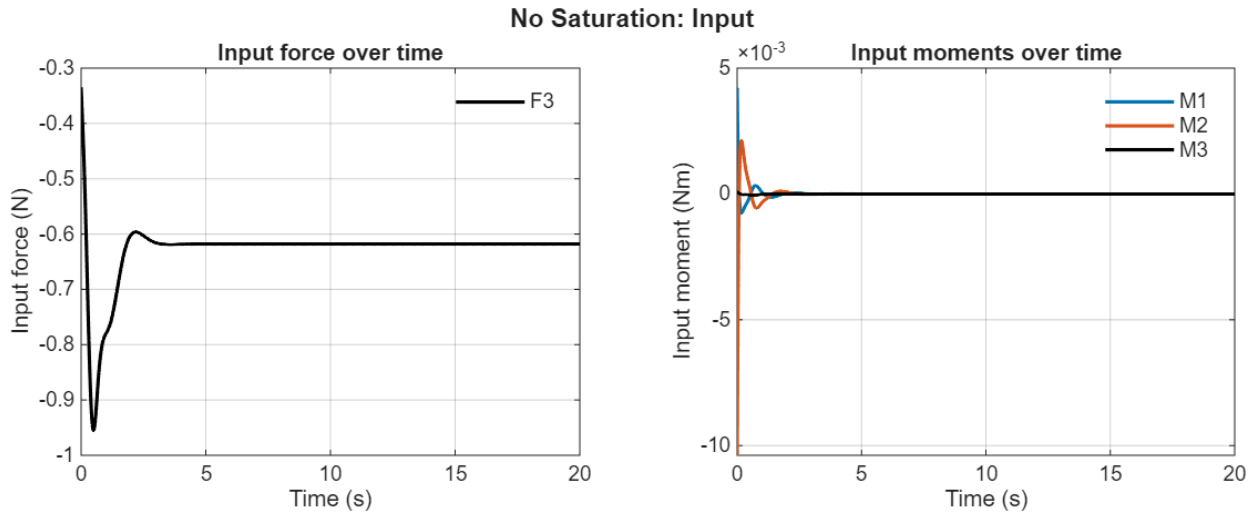


Figure 42: Input without saturation

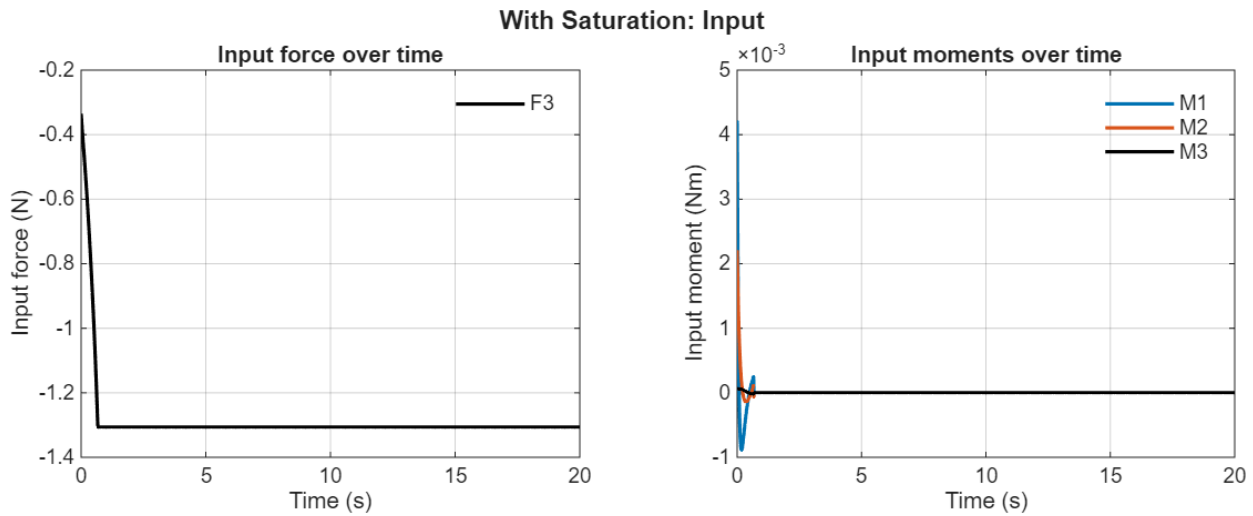


Figure 43: Input with saturation

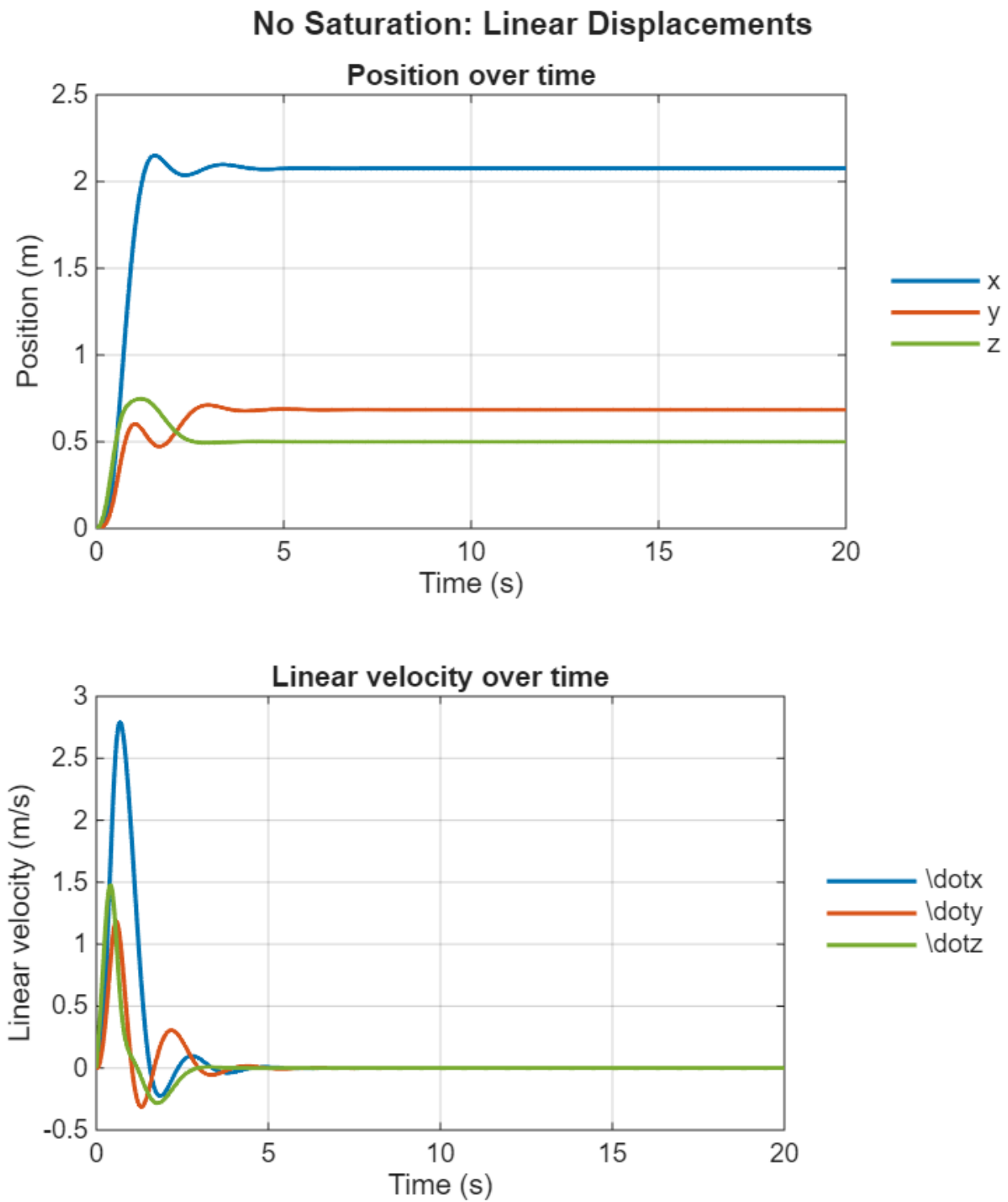


Figure 44: Linear displacements when there isn't saturation

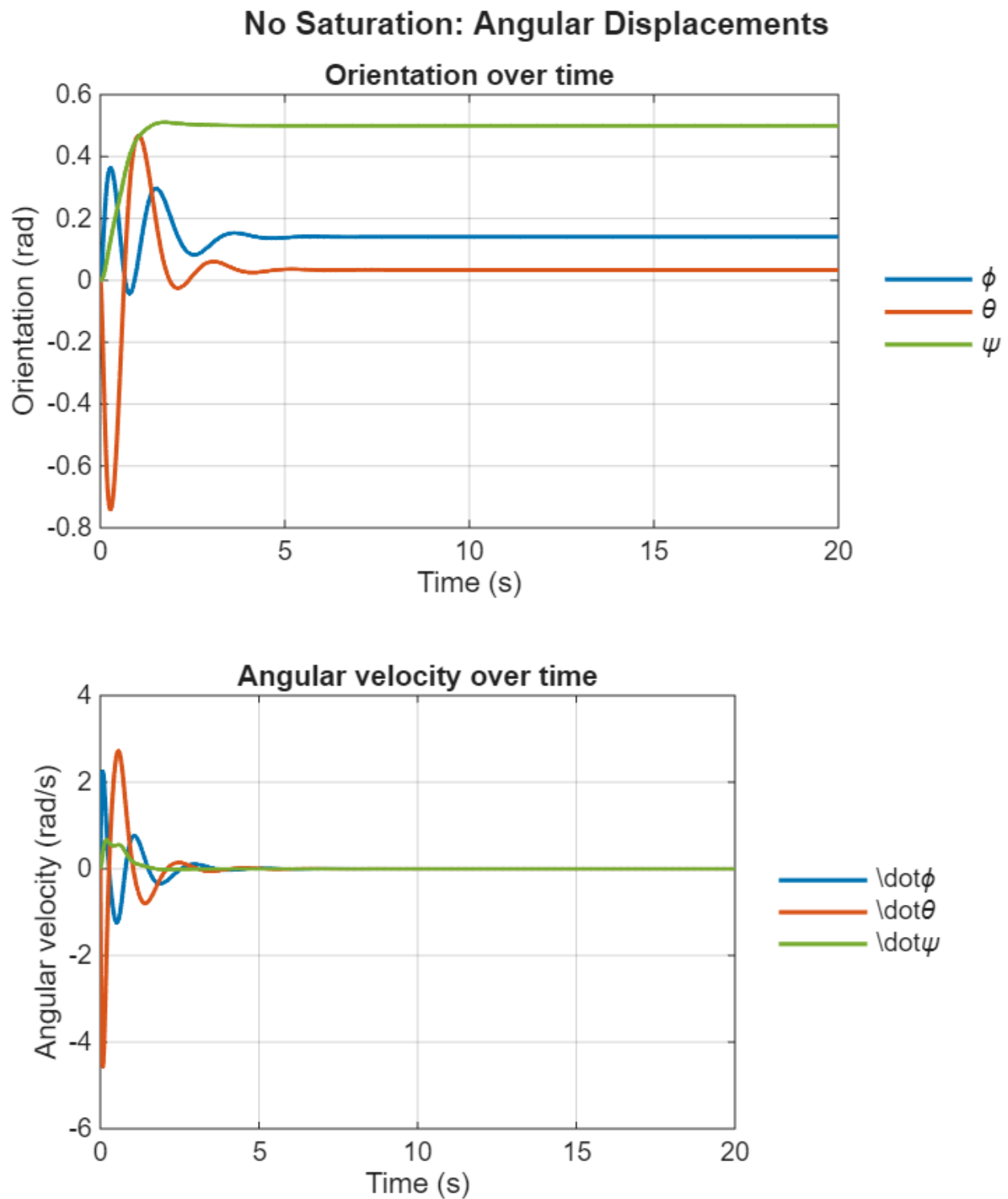


Figure 45: Angular displacements when there isn't saturation

With Saturation: Linear Displacements (UNSTABLE)

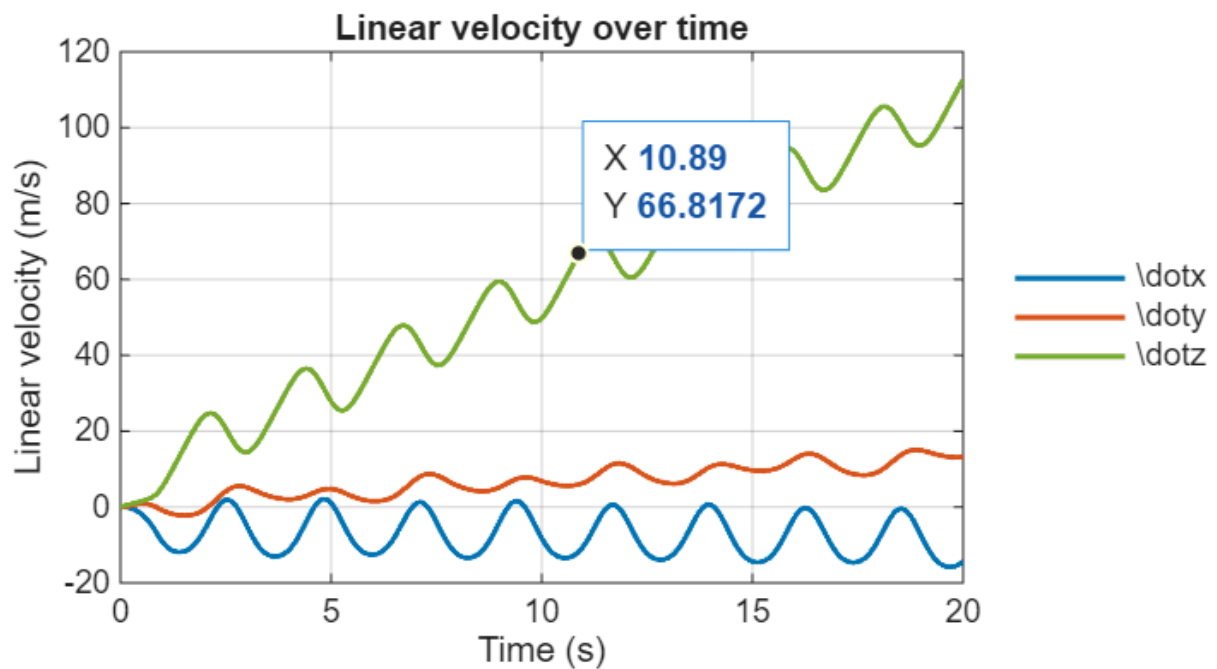
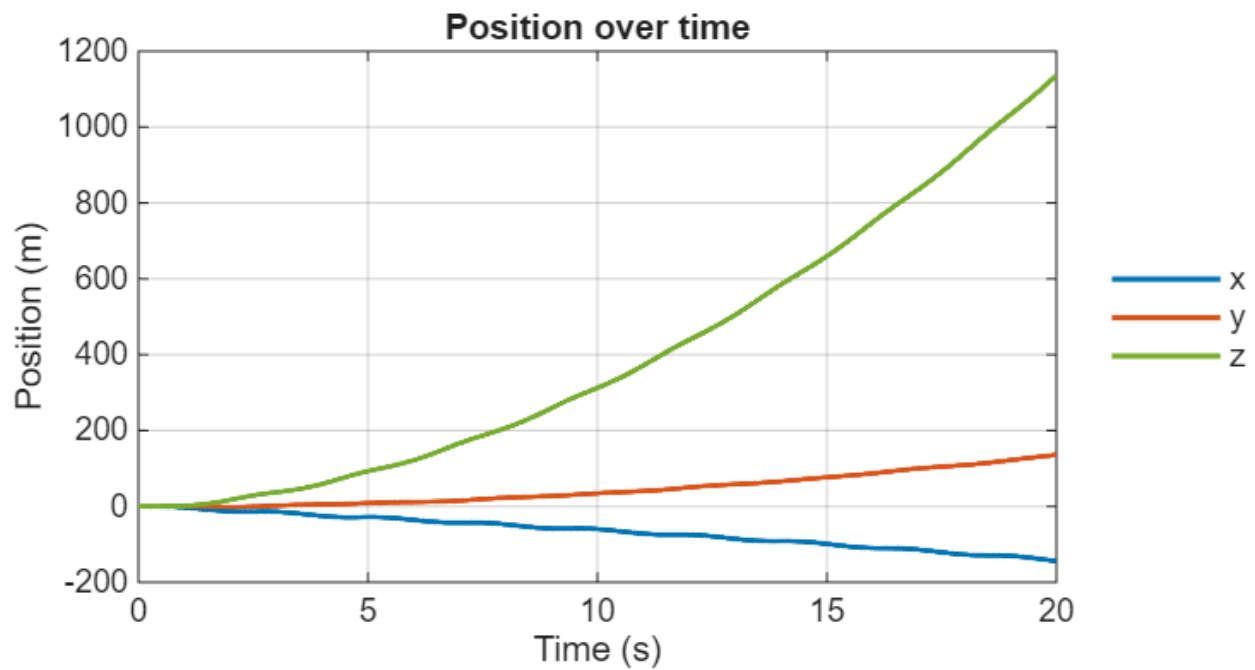


Figure 46: Linear displacements when there is saturation

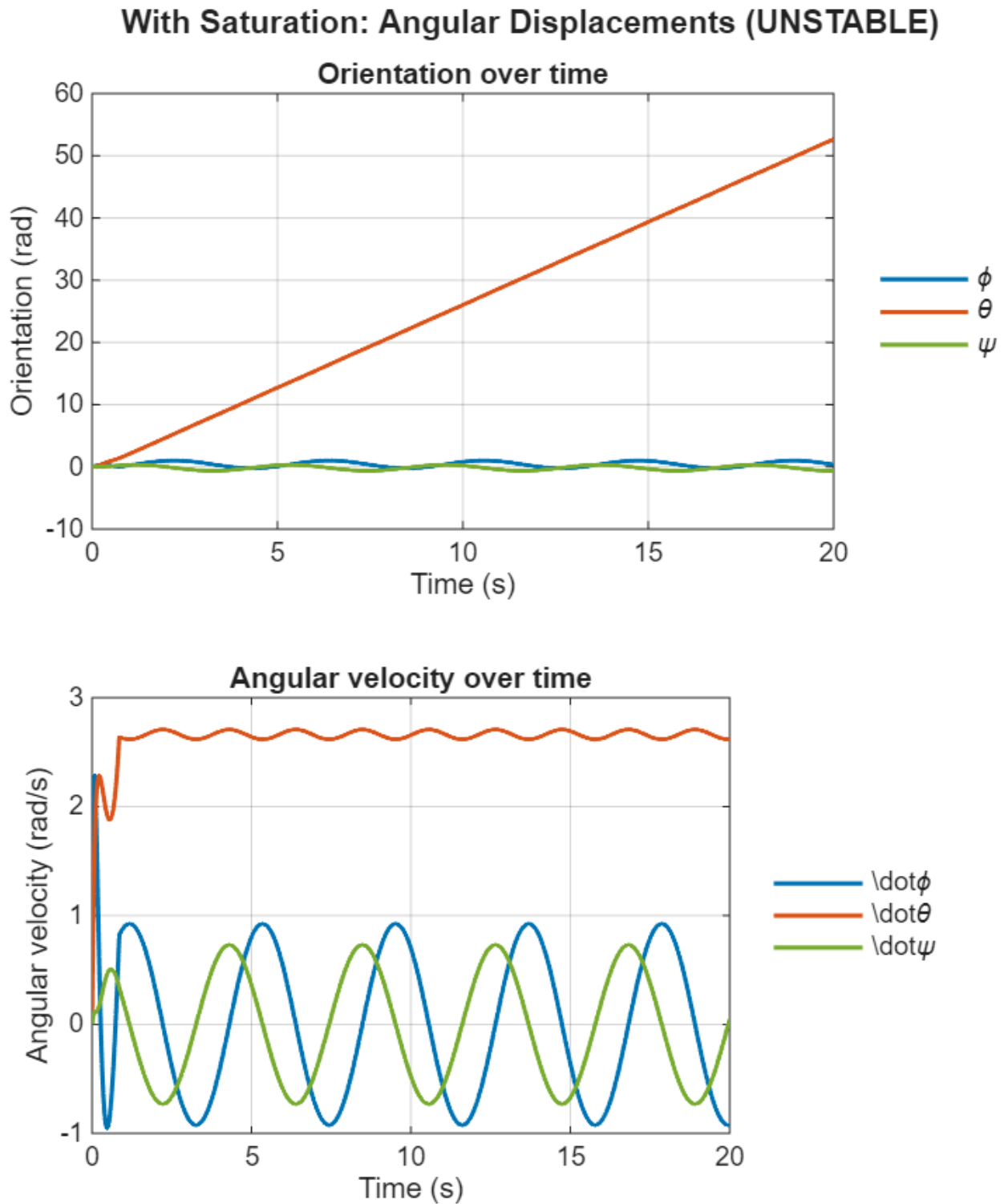


Figure 47: Angular displacements when there is saturation

Conclusion

This report covered the modeling and validation of the Parrot Mambo Fly quadcopter dynamics in Phase I of the laboratory project. The nonlinear model was implemented and validated, reproducing the expected behavior in hover,

step response tests, and the takeoff, rotate and land trajectory. The linearized model was then derived around the hovering equilibrium, resulting in four decoupled subsystems that were shown to be unstable, fully controllable and fully observable. Comparing the two models revealed the main limitations of linearization — the absence of Coriolis coupling and the small angle assumption — which become relevant as the drone moves away from hover. A pole-placement regulator was designed to stabilize the system, and the impact of motor saturation on closed-loop performance was evaluated.

The full-state feedback regulator designed using the Ackermann's pole-placement formula shows a successful stabilization of the quadcopter in both the linear and nonlinear models across all the conditions tested. The dominant complex pair with $\zeta = 0.7$ and $\omega_n = 3 \text{ rad/s}$ provides the best compromise between settling speed and overshoot. Deviations between the two models become visible only for under damped designs or large multi-axis perturbations, where nonlinear coupling effects dominate.

Motor saturation limits the achievable closed-loop performance, as pole placements that demand control effort beyond the actuator capacity will finish to destabilize the system. Therefore, any practical controller design must have in consideration the physical capacity of the motors, ensuring that the requested inputs remain between the admissible range on all the operating conditions.